从梯度下降到共轭梯度

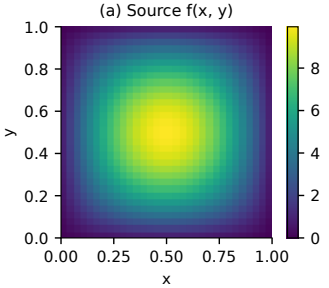
条件数、预条件与稀疏线性系统

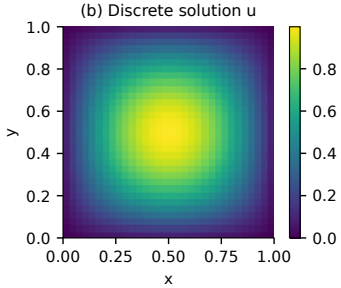
SKILL 应用测试

局部方程 → 能量几何 → 共轭方向 → 结构化预条件

从一张稳态场，得到一个线性系统

单位方形内求解 $-u_{xx} - \varepsilon u_{yy} = f$，边界 $u = 0$。u 是待求场，f 是源项，$\varepsilon = 0.01$ 表示纵向耦合较弱。





用已知解验证离散过程

选 $u_*(x, y) = \sin(\pi x)\sin(\pi y)$，

则 $f = (1 + \varepsilon)\pi^2 u_*$。

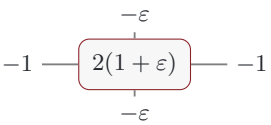
把场值放到网格上

每方向 32 个内部点，共 1024 个未知量。离散解与 u_* 的最大差约为 7.54×10^{-4}。

接下来把微分算子写成邻点间的代数关系，利用这套局部结构求解全部未知量。

一个五点模板，连接上千个未知量

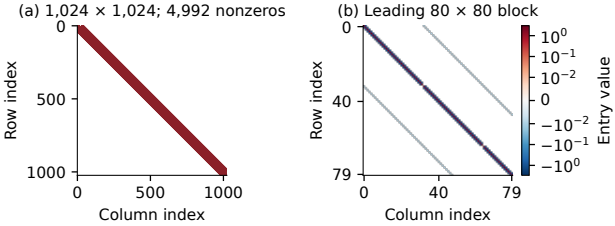
在单位方形求解 $-u_{xx} - \varepsilon u_{yy} = f$，边界 $u = 0$。每方向有 m 个内部点，$h = 1/(m+1)$。



$$L = I \otimes T + \varepsilon T \otimes I,$$

$$T = \mathrm{tridiag}(-1, 2, -1).$$

$\otimes$ 为 Kronecker 积，复制一维耦合。网格值按行展开为 x，有 $Lx = b$，$b = h^2 f$。



$m = 32$，$\varepsilon = 0.01$。

整体模式与左上角局部；按行编号后，非零元分布在五条带上。

稀疏结构直接决定存储与乘法成本

$n = m^2$，$\mathrm{nnz} = 5m^2 - 4m$；$m = 64$ 时 $n = 4096$，$\mathrm{nnz} = 20224$，乘法为 $O(\mathrm{nnz})$。

CSR 只存非零值、列索引和行指针；float64/32 位索引约 0.247 MiB，稠密存储为 128 MiB。

解方程等价于最小化二次能量

对称正定（SPD）：$A = A^{\mathsf{T}}$，且任意非零 v 满足 $v^{\mathsf{T}}Av > 0$。

$$f(x) = \tfrac{1}{2}x^{\mathsf{T}}Ax - b^{\mathsf{T}}x, \qquad \nabla f(x) = Ax - b.$$

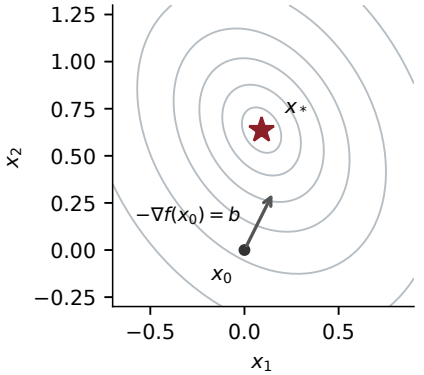
唯一极小点 x_* 满足 $Ax_* = b$。残差 $r = b - Ax$ 是负梯度；误差 $e = x_* - x$ 表示距真解的差。

把“离解多远”写成能量

$$\|e\|_A = \sqrt{e^{\mathsf{T}}Ae}, \qquad f(x) - f(x_*) = \tfrac{1}{2}\,\|e\|_A^2 .$$

等高线的轴向由特征向量决定，曲率由特征值决定。

$$A = \begin{bmatrix} 4 & 1 \\ 1 & 3 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad x_* = \begin{bmatrix} 1/11 \\ 7/11 \end{bmatrix}.$$



两条主轴的曲率约为 4.62 与 2.38；较大特征值对应更陡的方向。

沿负梯度走，步长由一维最小化确定

最速下降沿 $r_k = b - Ax_k$ 更新。精确线搜索在这条直线上选择能量最低的位置，下面记作 GD。

$$\phi(\alpha) = f(x_k + \alpha r_k).$$

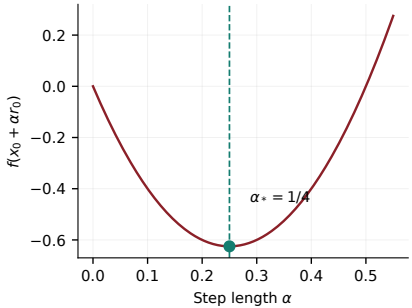
$$\begin{aligned}\phi'(\alpha) &= r_k^{\mathsf{T}}(Ax_k - b + \alpha A r_k) \\ &= -r_k^{\mathsf{T}} r_k + \alpha r_k^{\mathsf{T}} A r_k.\end{aligned}$$

令 $\phi'(\alpha_k) = 0$，得到

$$\alpha_k = \frac{r_k^{\mathsf{T}} r_k}{r_k^{\mathsf{T}} A r_k}, \qquad x_{k+1} = x_k + \alpha_k r_k.$$

本轮结束后，残差与刚走的方向正交

$r_{k+1} = r_k - \alpha_k A r_k$，因此
$r_{k+1}^{\mathsf{T}} r_k = 0$。

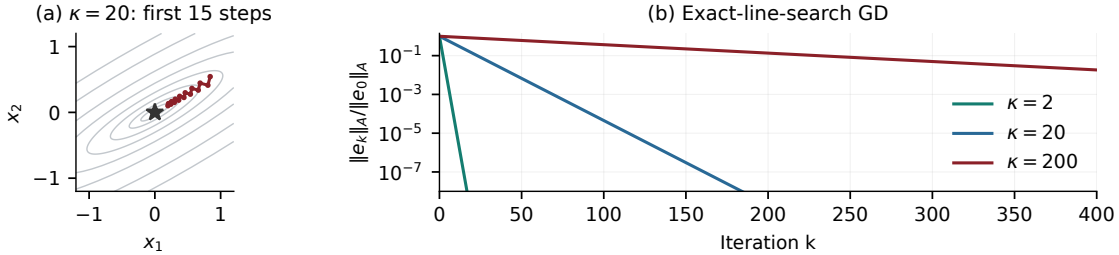


第 4 页的二维系统取 $x_0 = 0$：
$r_0 = (1, 2)^{\mathsf{T}}$，$r_0^{\mathsf{T}} A r_0 = 20$。
所以 $\alpha_0 = 5/20 = 1/4$，
$x_1 = (1/4, 1/2)^{\mathsf{T}}$。

条件数控制最速下降的收敛尺度

精确线搜索的梯度下降（GD）取 $x_{k+1} = x_k + \alpha_k r_k$。沿残差方向令导数为零：

$$\alpha_k = \frac{r_k^{\mathsf{T}} r_k}{r_k^{\mathsf{T}} A r_k}, \qquad \kappa = \frac{\lambda_{\max}}{\lambda_{\min}}, \qquad \frac{\|e_k\|_A}{\|e_0\|_A} \le \left(\frac{\kappa - 1}{\kappa + 1}\right)^k.$$



右图取 $A = \mathrm{diag}(1, \kappa)$、$x_* = 0$、$x_0 = (1, 1/\kappa)^{\mathsf{T}}$，达到上述速率。曲率差异让迭代在狭长谷底反复转向。

左图将 $\kappa = 20$ 的系统旋转 30°，展示前 15 步路径。

把“方向正交”改写成能量几何中的共轭

沿方向 p 完成一次精确线搜索后，有 $p^{\mathsf{T}}\nabla f(x)=0$。再沿 q 移动，梯度改变 αAq。

保持已经完成的最优条件

$$p^{\mathsf{T}}\nabla f(x+\alpha q)=\underbrace{p^{\mathsf{T}}\nabla f(x)}_{0}+\alpha p^{\mathsf{T}}Aq.$$

选择 $p^{\mathsf{T}}Aq=0$，新一步保留旧方向上的最优条件。这样的两方向称为A 共轭。

$$\langle p,q\rangle_A=p^{\mathsf{T}}Aq.$$

这把矩阵定义的曲率纳入方向之间的关系。

共轭梯度（CG）逐步构造这些方向，把每次搜索积累为对整个子空间的优化。

二维实例：普通直角与共轭的区别

仍取 $A=\left[\begin{smallmatrix}4&1\\1&3\end{smallmatrix}\right]$。坐标单位向量 $e_1=(1,0)^{\mathsf{T}}$、$e_2=(0,1)^{\mathsf{T}}$ 满足

$$e_1^{\mathsf{T}}e_2=0,\qquad e_1^{\mathsf{T}}Ae_2=1.$$

改取 $p=(1,2)^{\mathsf{T}}$、$q=(-7,6)^{\mathsf{T}}$：

$$Aq=(-22,11)^{\mathsf{T}},\quad p^{\mathsf{T}}Aq=0.$$

换到能量坐标，关系变得直观

$A^{1/2}$ 为 A 的对称正定平方根。

$u=A^{1/2}p$，$v=A^{1/2}q$，则

$u^{\mathsf{T}}v=p^{\mathsf{T}}Aq=0$。

共轭方向在能量坐标中构成普通直角。

共轭方向保留已经完成的修正

当 $p_i^{\mathsf{T}} A p_j = 0$、$i \neq j$，两方向为$A$ 共轭。沿新方向优化后，先前各方向的最优条件仍保持。

$$A = \begin{bmatrix} 4 & 1 \\ 1 & 3 \end{bmatrix}, \quad b = (1,2)^{\mathsf{T}}, \quad x_0 = 0,\ p_0 = r_0 = b.$$

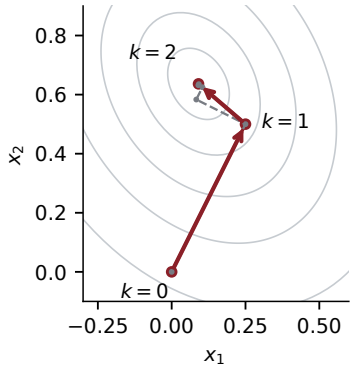
$$\alpha_0 = \tfrac{1}{4}, \qquad x_1 = (\tfrac{1}{4}, \tfrac{1}{2})^{\mathsf{T}},$$

$$r_1 = (-\tfrac{1}{2}, \tfrac{1}{4})^{\mathsf{T}}, \qquad \beta_0 = \frac{r_1^{\mathsf{T}} r_1}{r_0^{\mathsf{T}} r_0} = \tfrac{1}{16},$$

$$p_1 = r_1 + \beta_0 p_0 = (-\tfrac{7}{16}, \tfrac{3}{8})^{\mathsf{T}}.$$

旧方向导数 $p_i^{\mathsf{T}} \nabla f$ 的变化量为

$$\Delta(p_i^{\mathsf{T}} \nabla f) = \alpha p_i^{\mathsf{T}} A p_j = 0.$$



红线：CG 两步　灰虚线：GD 前五步。

第二步到达精确解

$p_0^{\mathsf{T}} A p_1 = 0$，$\alpha_1 = 4/11$，因此 $x_2 = x_1 + \alpha_1 p_1 = (1/11, 7/11)^{\mathsf{T}} = x_*$。

Krylov 子空间把矩阵乘法变成搜索空间

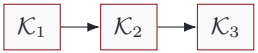
初始残差 r_0 与它反复乘 A 得到的向量，构成一个逐步扩展的搜索空间：

$$\mathcal{K}_k(A, r_0) = \operatorname{span}\{r_0, Ar_0, \dots, A^{k-1} r_0\}.$$

在已建立的空间中选择最优点

$$x_k = \arg\min_{x \in x_0 + \mathcal{K}_k} f(x).$$

等价地，$\|x_* - x_k\|_A$ 最小，
r_k 与已搜索的子空间正交。



每次矩阵乘法提供新的候选方向，短递推维护搜索方向与残差。

二维实例在第二步覆盖整个空间

$$r_0 = (1, 2)^{\mathsf{T}}, \quad Ar_0 = (6, 7)^{\mathsf{T}}.$$

两列行列式为 -5，故 $\mathcal{K}_2 = \mathbb{R}^2$。

$$x_* = \frac{7}{11} r_0 - \frac{1}{11} Ar_0 = (1/11, 7/11)^{\mathsf{T}}.$$

精确解已经属于第二步的搜索空间，因此第 8 页的两次线搜索即可到达 x_*。

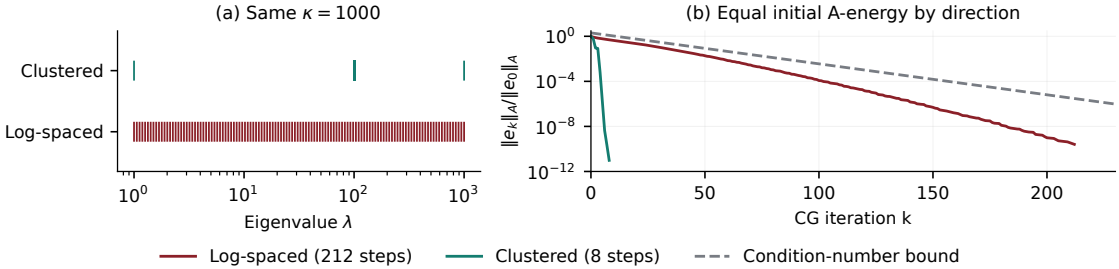
误差多项式给出另一种读法

$e_k = \pi_k(A) e_0$，$\pi_k(0) = 1$。本例 $\pi_2(t) = 1 - \frac{7}{11} t + \frac{1}{11} t^2$，
在两个特征值处均为零，两个特征方向的误差同时消失。

谱聚集可以大幅加速 CG

谱指 A 的特征值分布。误差多项式 π_k 在每个特征值处缩放对应方向的误差；特征值聚集时，同一个低阶多项式可以同时压低一整簇分量：

$$e_k = \pi_k(A)e_0, \quad \pi_k(0) = 1, \qquad \frac{\|e_k\|_A}{\|e_0\|_A} \le 2\left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\right)^k.$$



120 维，$x_0 = 0$、$(x_*)_i = \lambda_i^{-1/2}$，各特征方向初始 A 能量相同。聚集谱含 $1, 1000$ 与 118 个位于 $[100, 101]$ 的值。float64 真相对残差达 10^{-10}：聚集谱 8 步，对数谱 212 步。

CG 用短递推保存方向搜索的成果

初始化 $r = b - Ax$、$p = r$。每轮沿 p 搜索，再用新残差与旧方向组成下一方向。

$$q_k = Ap_k, \qquad \alpha_k = \frac{r_k^{\mathsf{T}} r_k}{p_k^{\mathsf{T}} q_k},$$

$$x_{k+1} = x_k + \alpha_k p_k,$$

$$r_{k+1} = r_k - \alpha_k q_k,$$

$$\beta_k = \frac{r_{k+1}^{\mathsf{T}} r_{k+1}}{r_k^{\mathsf{T}} r_k},$$

$$p_{k+1} = r_{k+1} + \beta_k p_k.$$

```
r = b - A @ x
p, rr = r.copy(), r @ r
target = 1e-8 * np.linalg.norm(b)
for k in range(5000):
    if np.sqrt(rr) <= target: break
    q = A @ p
    alpha = rr / (p @ q)
    x += alpha * p
    r -= alpha * q
    rr1 = r @ r
    p = r + (rr1 / rr) * p
    rr = rr1
true_residual = b - A @ x
```

从核心运算核算成本

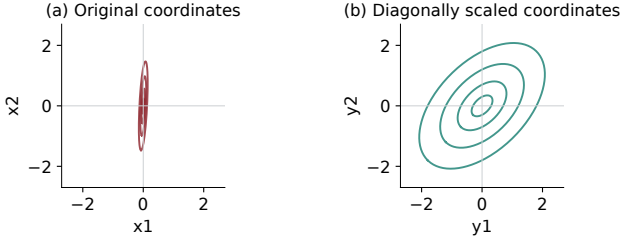
每步 1 次稀疏矩阵向量乘法（SpMV）、2 次内积及向量更新，存储少量 n 维向量。

A, b, x 为系统与初值，np 为 NumPy；最后重算真残差，按第 18 页的标准检查结果。

预条件通过改变坐标，改善能量几何

选易解的 SPD 矩阵 $P = CC^{\mathsf{T}}$，令 $y = C^{\mathsf{T}}x$，得到

$$\widetilde{A}y = C^{-1}b, \qquad \widetilde{A} = C^{-1}AC^{-\mathsf{T}}.$$



$A = \left[\begin{smallmatrix} 200 & -10 \\ -10 & 2 \end{smallmatrix}\right]$，$P = \mathrm{diag}(200, 2)$。
条件数由约 134 降至 3，等能量曲线更均衡。

预条件共轭梯度（PCG）在原坐标中实现这一过程，每轮计算 $z = P^{-1}r$。
收敛界中的条件数变为 $\kappa(\widetilde{A})$。

选择 P 时同时考虑几何改善与求解成本：越贴近 A 的结构，越需要高效实现它的逆作用。

预条件同时处理尺度与耦合

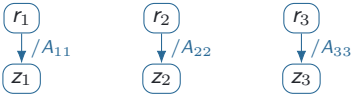
令 $P = CC^{\mathsf{T}}$、$y = C^{\mathsf{T}}x$，等价系统为 $\widetilde{A}y = C^{-1}b$，$\widetilde{A} = C^{-1}AC^{-\mathsf{T}}$。$\widetilde{A}$ 保持 SPD，PCG 收敛界使用 $\kappa(\widetilde{A})$。

Jacobi：对角均衡，消除变量尺度

对第 3 页的 L 做对称缩放，构造 $A = SLS$，$\varepsilon = 0.01$。$S = \mathrm{diag}(s)$，s 沿每行由 1 几何增至 20。

$$P_J = \mathrm{diag}(A), \qquad z_i = r_i/(P_J)_{ii}.$$

$m = 32$：$\kappa(A) = 16895.05$，
对角均衡后 $\kappa_J = 440.69$。



各分量分别除以对应对角元

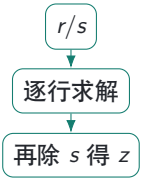
Line：行块预条件，保留强耦合

$$P_B = S[I \otimes (T + 0.02I)]S.$$

每行求解同一个三对角系统，长度 m 的 Cholesky 因子复用于全部 m 行。
$m = 32$：$\kappa_B = 5.35$。



按强 x 耦合分行



实际计算的是 $P^{-1}r$。SciPy 的参数 `M` 接受这个逆作用。代码中的 `apply_Pinv` 即对应操作。

行块预条件：一个带状因子服务所有网格行

$P_B = S[I \otimes B]S$，其中 $B = T + 2\varepsilon I$。先解尺度、再解每行三对角系统、最后恢复尺度：

$$P_B^{-1} r = S^{-1}[I \otimes B^{-1}]S^{-1} r.$$

```
from scipy.linalg import (
    cholesky_banded, cho_solve_banded)
band = np.zeros((2, m))
band[0] = 2 + 2 * eps
band[1, :-1] = -1
chol = cholesky_banded(band, lower=True)
def apply_Pinv(r):
    U = (r / s).reshape(m, m).T
    Z = cho_solve_banded((chol, True), U)
    return Z.T.ravel() / s
```

带状存储与批量求解

band[0] 存对角，band[1] 存次对角。转置后，每一列是一个网格行的右端。

PCG 对 CG 递推的修改

先求 $z = P^{-1} r$，初始方向 $p = z$，令

$\rho = r^{\mathsf{T}} z$。

$\alpha = \rho / (p^{\mathsf{T}} A p)$，

$\beta = \rho_{\text{new}} / \rho$，

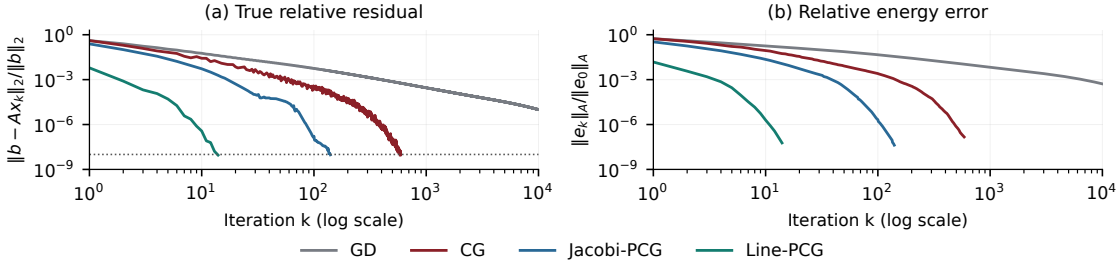
$p_{\text{new}} = z_{\text{new}} + \beta p$。

分解长度 m 的 B 后复用因子，每轮解 m 个右端，整体工作为 $O(m^2) = O(n)$。

m 为每行内部点数，s 为第 13 页的缩放向量。

同一系统上的四种迭代轨迹

$m = 32,\ n = 1024,\ A = SLS,\ \varepsilon = 0.01,\ s_i \in [1, 20]$。取 $x_* \sim \mathcal{N}(0, I)$（种子 20261111）、$b = Ax_*$、$x_0 = 0$，使用 float64。

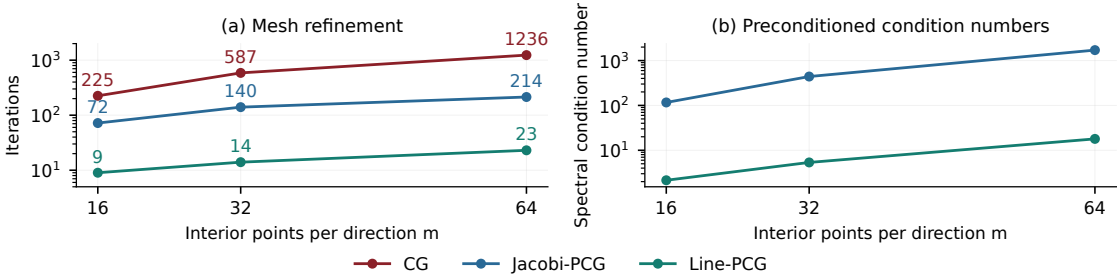


方法	GD	CG	Jacobi-PCG	Line-PCG
达到真相对残差 10^{-8} 的步数	> 10000	587	140	14
终止时真相对残差	1.01×10^{-5}	9.26×10^{-9}	9.49×10^{-9}	9.05×10^{-9}

GD 截至 10000 步。图中 $e_k = x_* - x_k$。行块保留强耦合，将 Jacobi-PCG 的 140 步降至 14 步。

网格加密后的收益与成本

$m = 16/32/64$ 对应 $n = 256/1024/4096$。保持 $\varepsilon = 0.01$、$s_i \in [1, 20]$ 与右端构造方法，全部按真相对残差 10^{-8} 停止。



每步增加多少工作

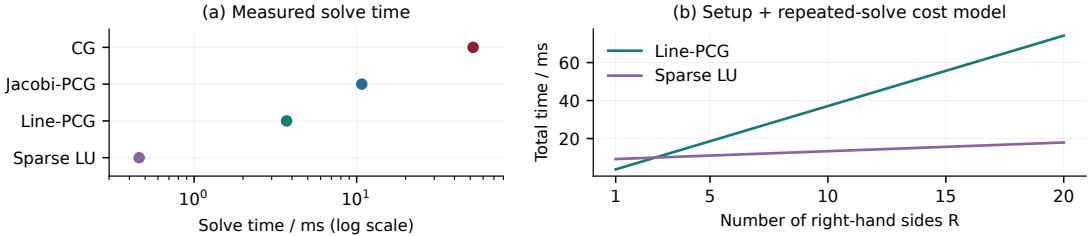
三方法都有 $O(\mathrm{nnz})$ 的 SpMV。Jacobi 加 n 次逐项除法；行块加 m 个长度 m 的三对角求解，总计 $O(n)$。

总成本如何判断

建预条件成本 + 迭代数 × 每步成本。行块的 κ_B 随网格加密由 2.14 升至 17.94；同一矩阵的多个右端可复用因子。

迭代数与每步成本，共同决定总时间

对 $m = 64$ 的同一系统，用 SciPy 1.18.1 在本机预热后重复 7 次，取中位数。求解阈值为 10^{-8}，时间单位为毫秒。



方法	CG	Jacobi-PCG	Line-PCG	稀疏 LU
准备	0.004	0.068	0.045	8.745
单次求解	52.066	10.752	3.706	0.459

右图按 $T(R) = T_{\text{setup}} + R\,T_{\text{solve}}$ 估算。行块减少迭代；同一矩阵的多个右端可以复用预条件或 LU 因子。

Windows 本机，线程环境设为 1。

用真残差停止，再核对误差尺度

SciPy 的 cg 接受 SPD 矩阵 A；算子 M 实现 P^{-1} 的作用。

用原方程检查停止条件

$$\|b - Ax\|_2 \le \max\{\texttt{rtol}\,\|b\|_2,\ \texttt{atol}\}.$$

返回后重算 $b - Ax$ 并检查 `info`；`rtol` 控制相对残差，`atol` 控制绝对残差。

用条件数换算误差尺度

$$\frac{\|e\|_2}{\|x_*\|_2} \le \kappa(A)\frac{\|r\|_2}{\|b\|_2}.$$

$\kappa \approx 1.69 \times 10^4$ 时，残差 10^{-8} 对应的通用相对误差上界约 1.69×10^{-4}。

```
import numpy as np, scipy.sparse.linalg as sla
M = sla.LinearOperator(A.shape, matvec=apply_Pinv)
x, info = sla.cg(A, b, M=M, rtol=1e-8, atol=0.0, maxiter=5000)
res = np.linalg.norm(b - A @ x)
assert info == 0 and res <= 1e-8 * np.linalg.norm(b)
```

行块实例（$m = 32$）：`info = 0`（收敛），$\|b - Ax\|_2 / \|b\|_2 = 9.05 \times 10^{-9}$。

精确算术下 CG 至多 n 步；舍入会损失共轭性，第 10 页 120 维例用了 212 步。用递推残差迭代，用真残差验收。

直接实现五点乘法，也能接入同一个 CG 接口

CG 只需要计算 Av。对 $A = SLS$，先缩放向量，在网格上执行邻点加权，再缩放回来。

```
def matvec(v):
    U = (s * v).reshape(m, m)
    V = (2 + 2*eps) * U.copy()
    V[:, :-1] -= U[:, 1:]
    V[:, 1:] -= U[:, :-1]
    V[:-1, :] -= eps * U[1:, :]
    V[1:, :] -= eps * U[:-1, :]
    return s * V.ravel()
Aop = sla.LinearOperator(
    (m*m, m*m), matvec=matvec,
    dtype=np.float64)
```

邻点边界直接反映零边界条件

切片只累加内部邻点；边界外 $u = 0$，对本页齐次边界无需增加右端项。

在 $m = 32$ 上做数值对照

同一测试向量，显式矩阵与算子乘法的相对差为 1.67×10^{-16}。

配行块预条件，真相对残差为 9.05×10^{-9}。

```
M = sla.LinearOperator(Aop.shape, matvec=apply_Pinv)
x, info = sla.cg(Aop, b, M=M, rtol=1e-8, atol=0)
res = np.linalg.norm(b - Aop @ x) / np.linalg.norm(b)
```

`sla` 为 `scipy.sparse.linalg`；矩阵自由实现主要存储向量及预条件数据。

把结构、收敛与成本连成一条求解路径

矩阵或任务特征	具体选择
对称正定且稀疏	CG / PCG，利用稀疏乘法与少量工作向量
变量尺度差异大	用对角预条件先做尺度均衡
沿某方向或变量组强耦合	按线或物理变量分块，保留局部耦合
同一矩阵、多个右端	复用预条件与因子，摊薄构建成本
对称不定 / 一般非对称	分别考虑 MINRES / GMRES

把三条线索连起来

条件数连接能量几何与收敛上界；共轭方向减少重复修正；预条件把矩阵结构转化为更少的迭代。

Shewchuk (1994), *An Introduction to the Conjugate Gradient Method.*

Barrett 等，*Templates for the Solution of Linear Systems*，§2.3.1、§3.2。

SciPy 官方接口：`cg`、`minres`、`gmres`。

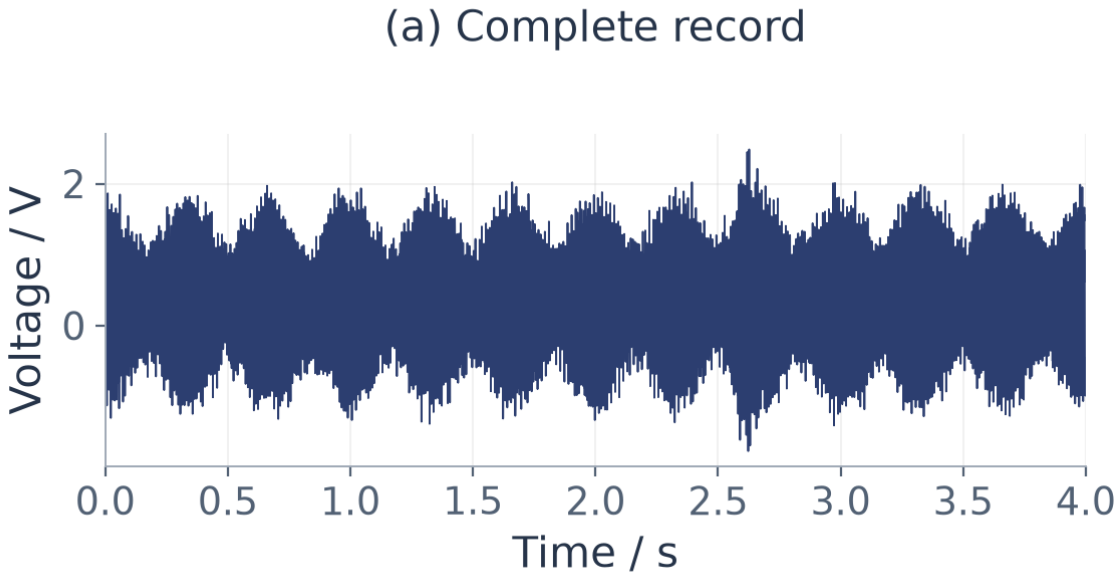
从传感器数据到可信频谱

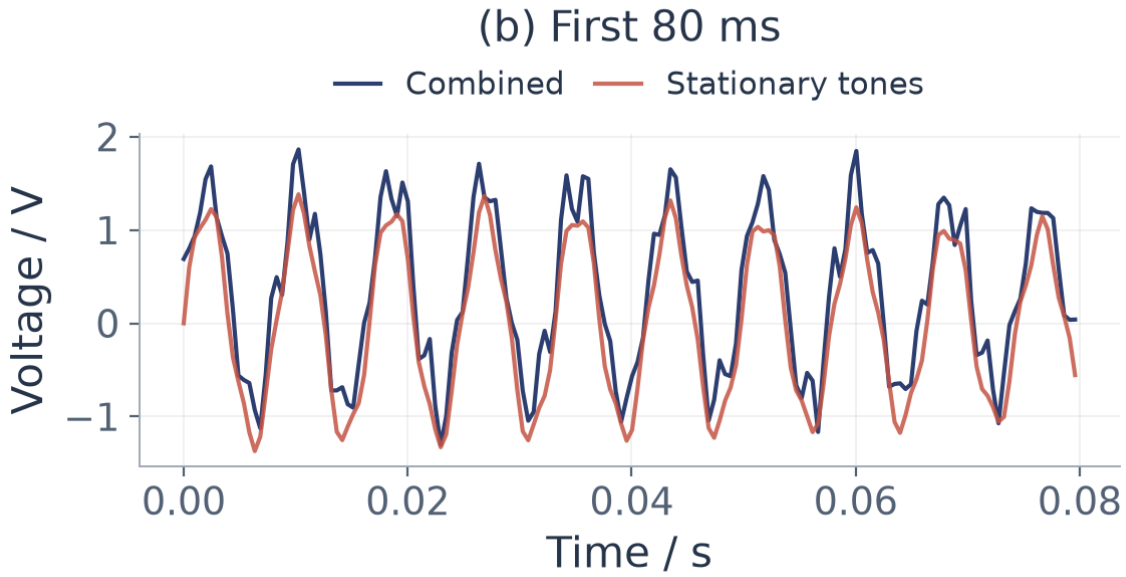
采样、FFT 与时频分析

SKILL应用测试

从可复算信号出发，连接数学、图表与分析代码。

先分清信号里有哪些变化





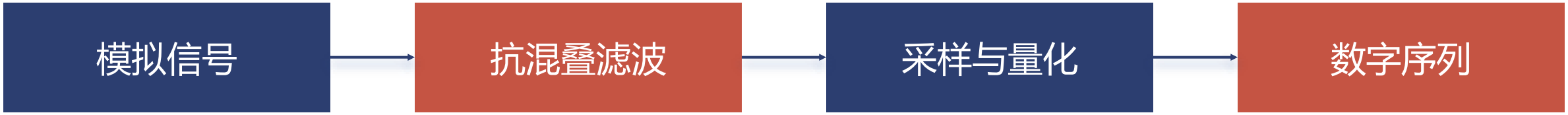
一段可复算的教学信号

稳定分量：120 / 123 / 420 Hz
扫频500→700 Hz；900 Hz短时事件
噪声标准差0.08 V；基线0.30→0.35 V
采样2048 Hz × 4 s = 8192点。

分析要回答的三个问题

哪些频率持续存在?
它们的幅值和功率是多少?
900 Hz 的扰动出现在什么时刻?

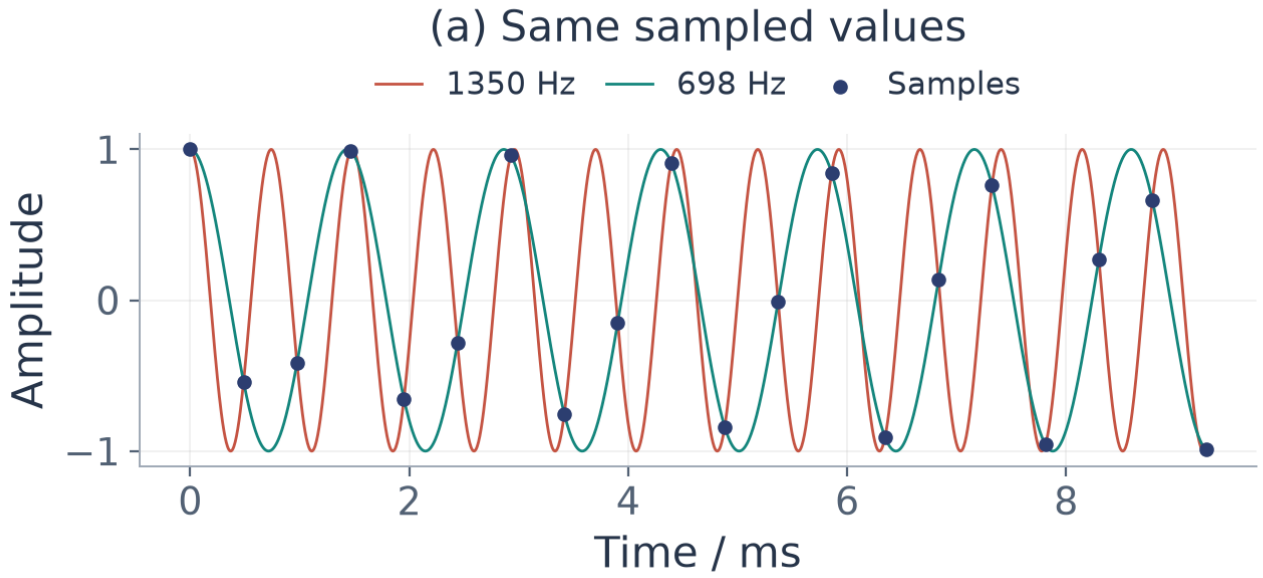
采样参数决定数据能回答什么

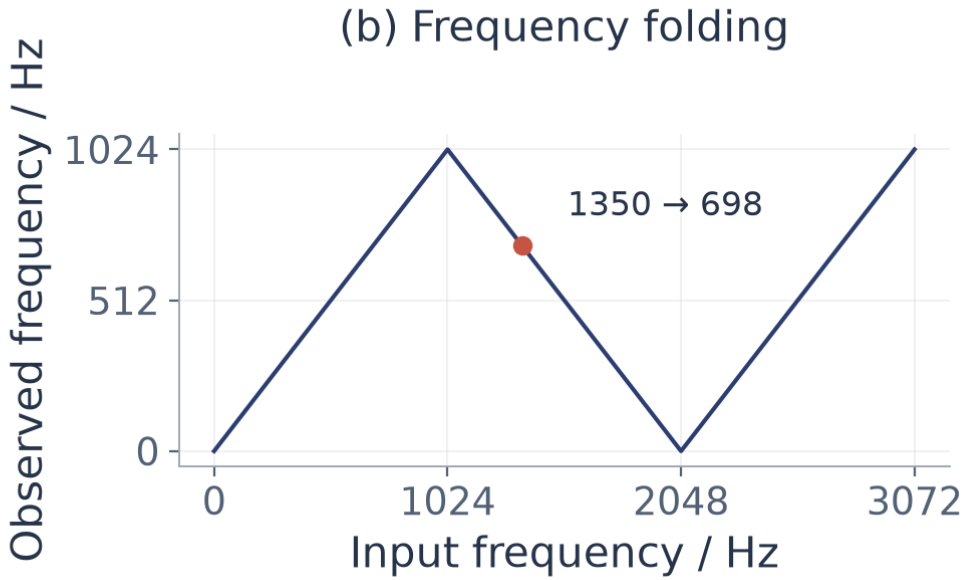


参数	本例设置	直接含义
采样率 f_s	2048 Hz	相邻采样相隔约 0.488 ms
记录时长 T	4 s	保留这段时间内的变化
样本数 N	8192	$N = f_s \times T$
频率间隔	0.25 Hz	$\Delta f = 1/T$

观测上限为 1024 Hz；目标频段与记录长度需要一起确定。

混叠让高频落到另一个位置





采样时刻的数学等价

f_s = 2048 Hz 时，1350 Hz 与 698 Hz 余弦在采样点上取值相同。

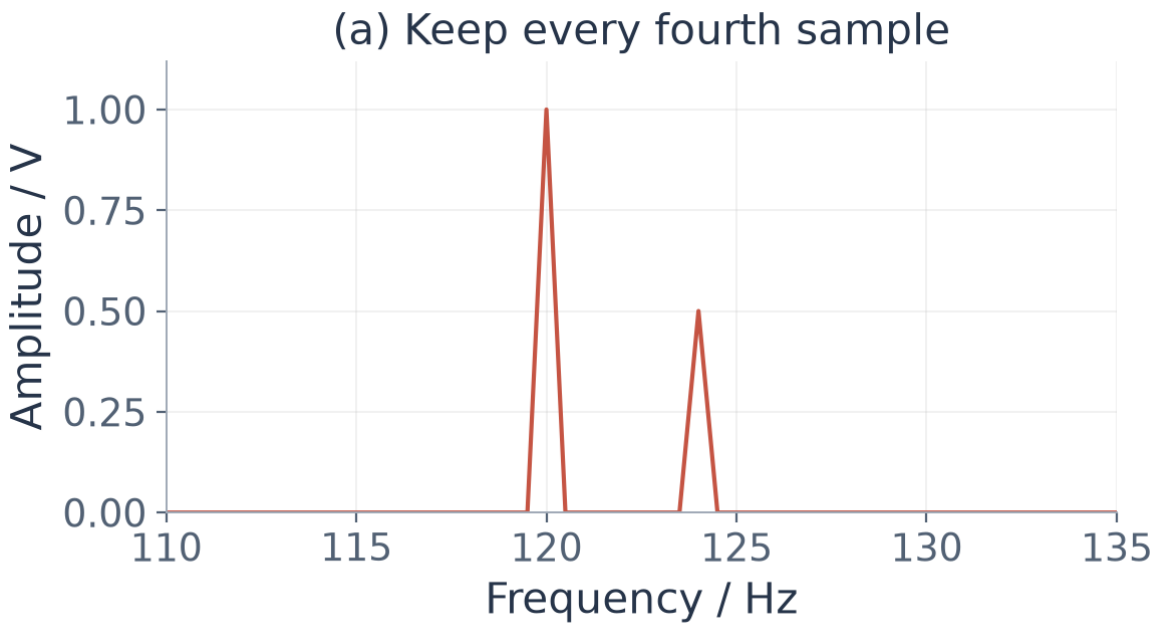
折叠频率取 $|f - kf_s|$，落入 0–$f_s/2$。

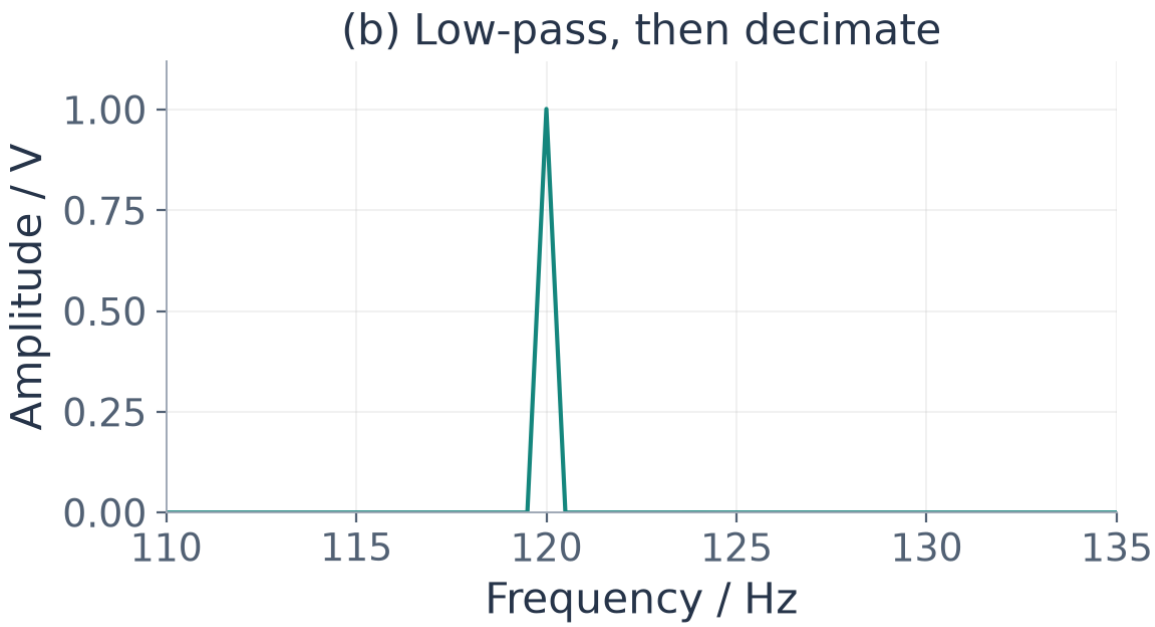
把边界落实到采集

连续信号先经过抗混叠滤波。

目标最高频率与滤波过渡带共同决定采样率，采样点本身不携带原频率标签。

降采样前先限定保留的频段





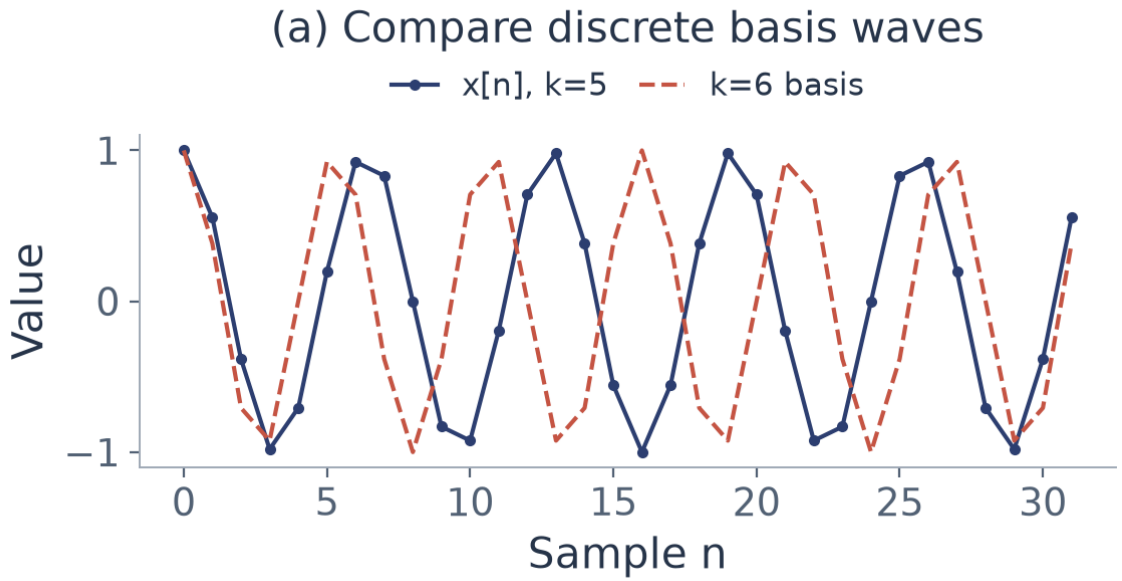
直接每隔 4 点取 1 点

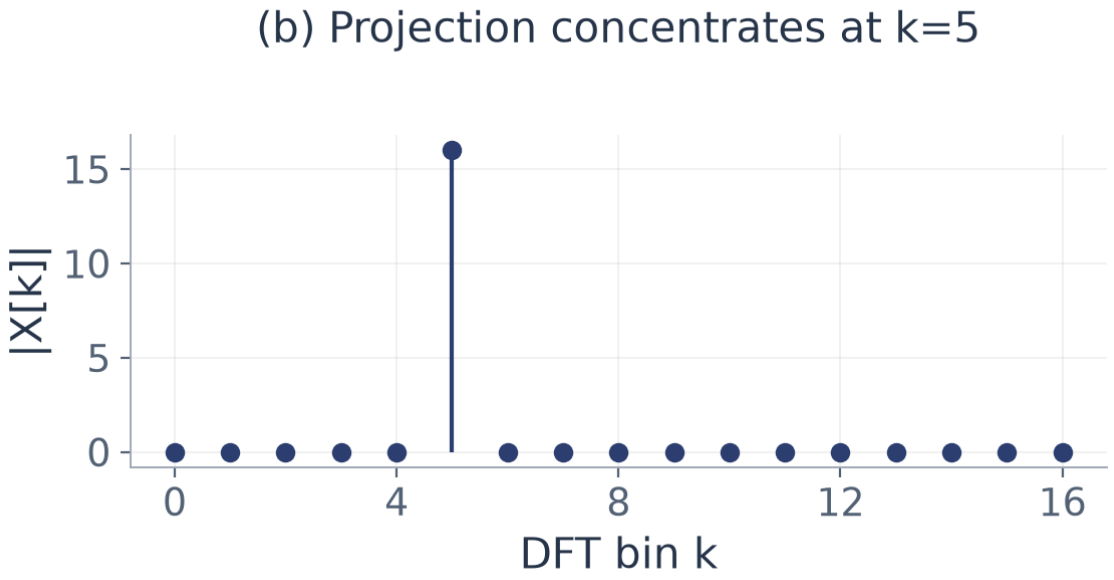
120 Hz 与 900 Hz 的双音信号，
降到 512 Hz 后，900 Hz 折叠至124 Hz；
假峰幅值为 0.500 V。

先低通，再降低采样率

resample_poly 处理后，124 Hz 分量
约为 5.69×10^{-5} V。
保留 900 Hz 事件时继续使用原采样率。

离散傅里叶变换：衡量频率基底的匹配





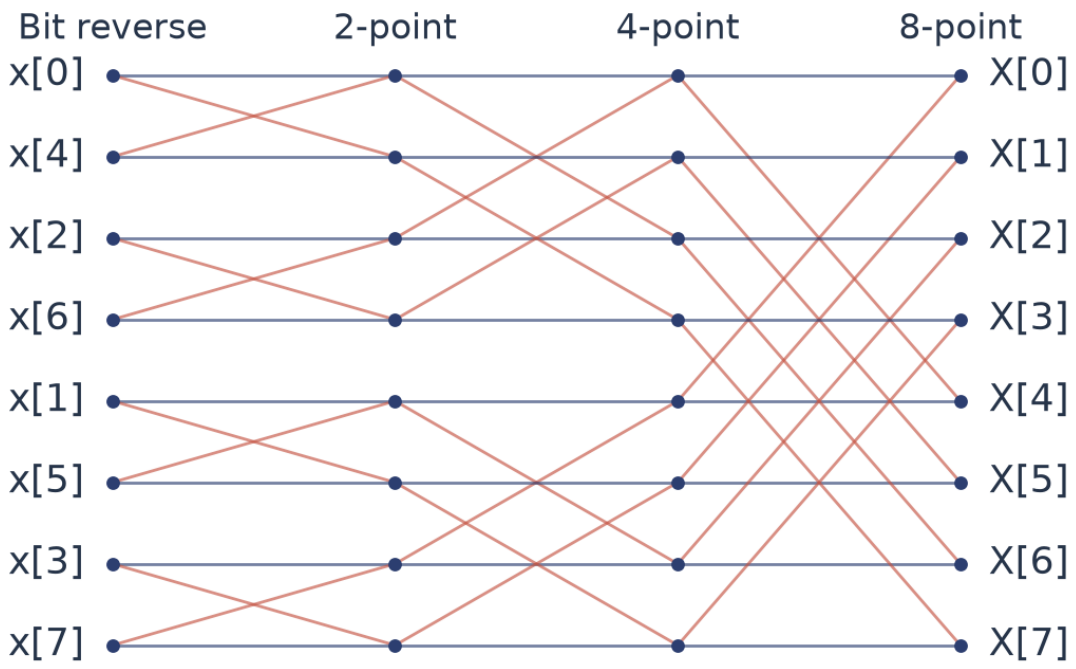
每个系数是一组加权求和

$X[k] = \Sigma_n\, x[n]\cdot\exp(-j2\pi kn/N)$
DFT复系数X[k]记录幅值与相位；
n=0,…,N−1，j为虚数单位。

频点与物理频率一一对应

频点编号k对应 $f_k = kf_s/N$；
32点中的5个周期，集中于k=5。
rfft（实数FFT）保留非负频率部分。

快速傅里叶变换：把 DFT 分解成蝶形



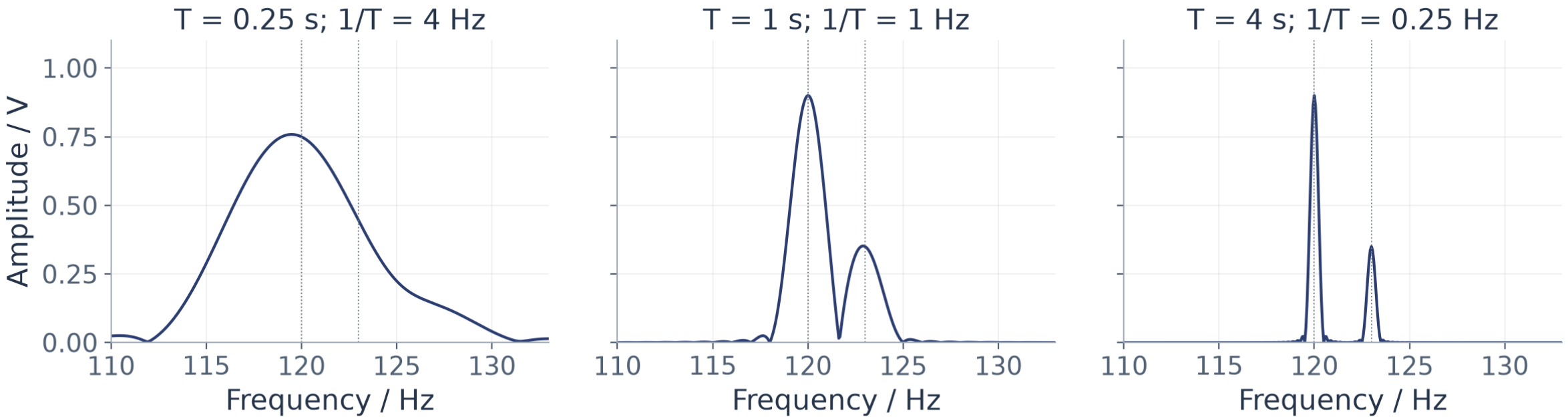
8点变换分为3级

输入先按位反转顺序排列。
u、v为蝶形输入；输出为
u + ωv 与 u − ωv，
ω为复数旋转因子。

N	直接复乘	蝶形数
1024	1,048,576	5,120
8192	67,108,864	53,248

每个蝶形包含1次复乘与2次复加；直接复乘按N^2计，含平凡乘法。
蝶形数为$N \cdot \log_2 N/2$；图示保留数据流，旋转因子由每一级决定。

延长记录，才能看清相近频率



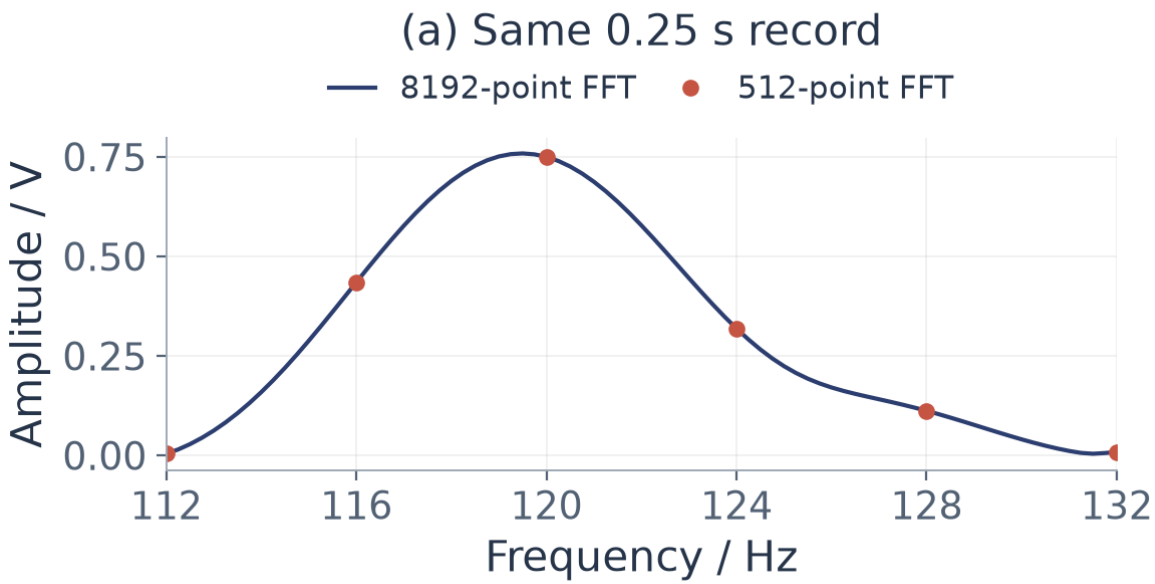
把观察时间作为分辨能力的来源

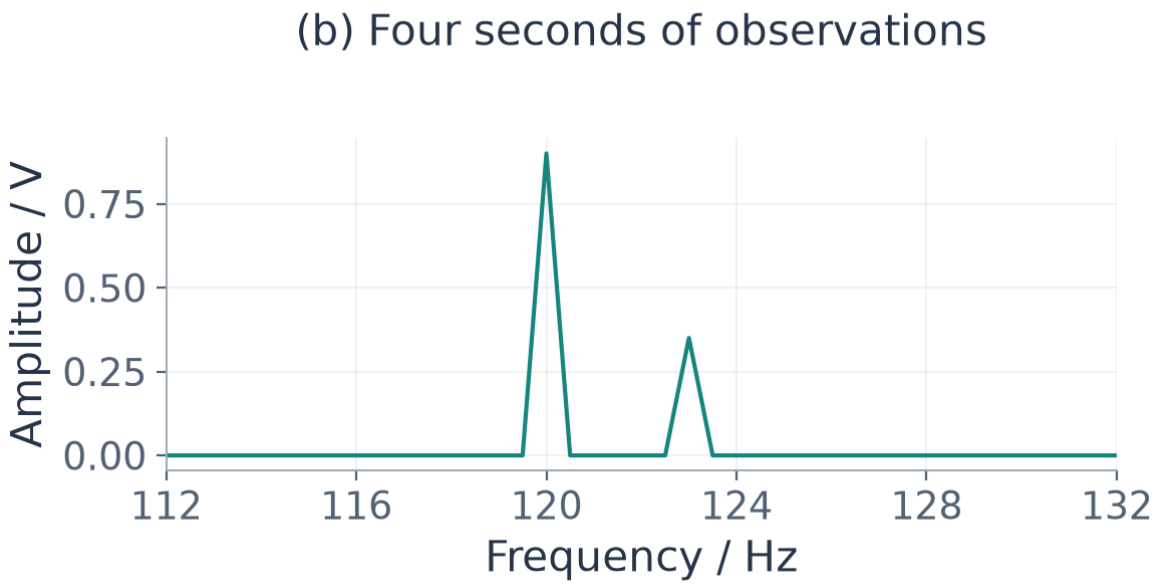
两条频率相差3 Hz，幅值为0.9/0.35 V。
Hann窗下，峰周围的主瓣随T增加变窄；
4 s记录中两个峰的位置与高度清楚。

把采样间隔与观测时间分开

频率栅格 $\Delta f = f_s/N = 1/T$。
固定时长提高采样率，会增加样本数；
延长时长提供区分相近频率的观察。

补零细化频谱采样，长记录收窄主瓣





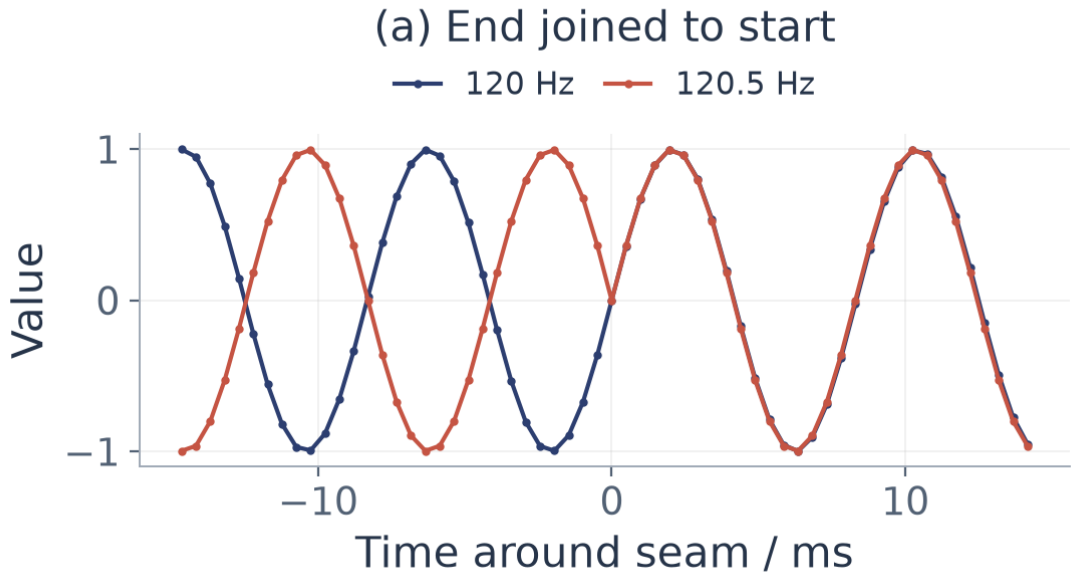
左图只改变 FFT 点数

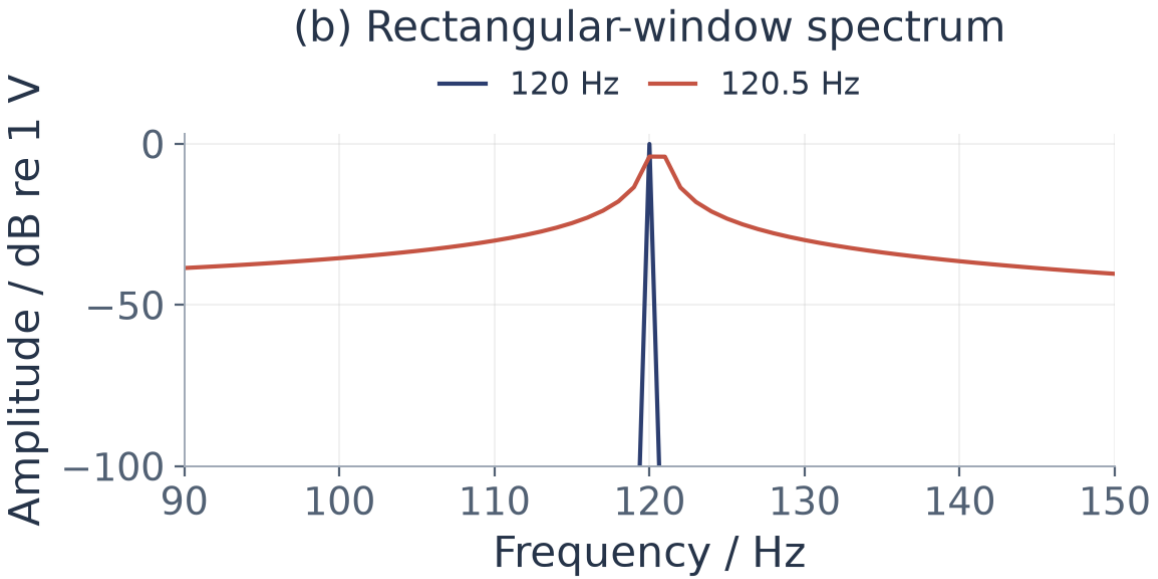
同一0.25 s数据，NFFT由512增至8192，
栅格从4 Hz细化到0.25 Hz；
曲线形状被更密的点描绘出来。

右图增加实际观察时长

4 s数据使两个峰分别集中于120/123 Hz。
选记录长度控制区分能力；
选补零长度改善读峰与插值显示。

有限记录的接缝带来频谱泄漏





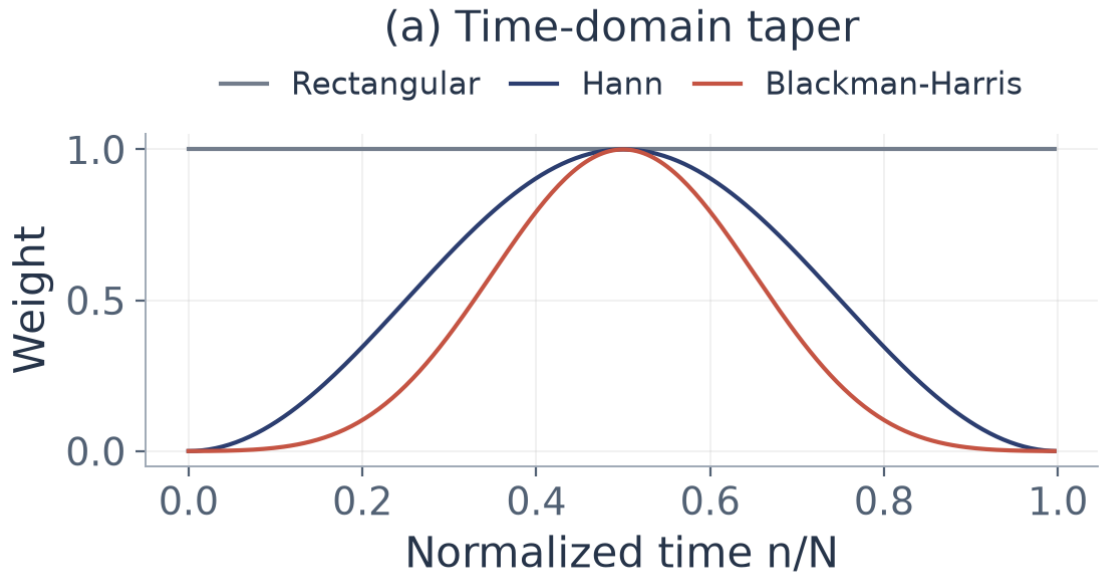
截取记录，相当于乘上时间窗

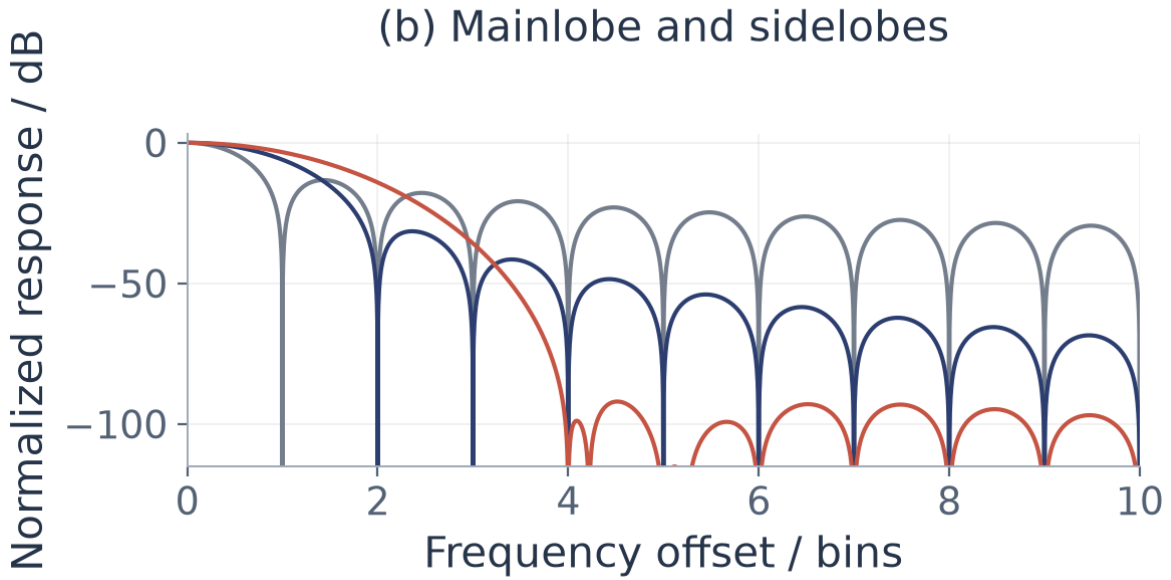
1 s内120 Hz恰有120个周期；
120.5 Hz首尾拼接后斜率突变。
边界变化将能量分布到更多频点。

先确定信号与窗口的组合

矩形窗保留每个样本的相同权重。
对非整周期分量，可通过平滑窗
降低远处旁瓣，再校准幅值和功率。

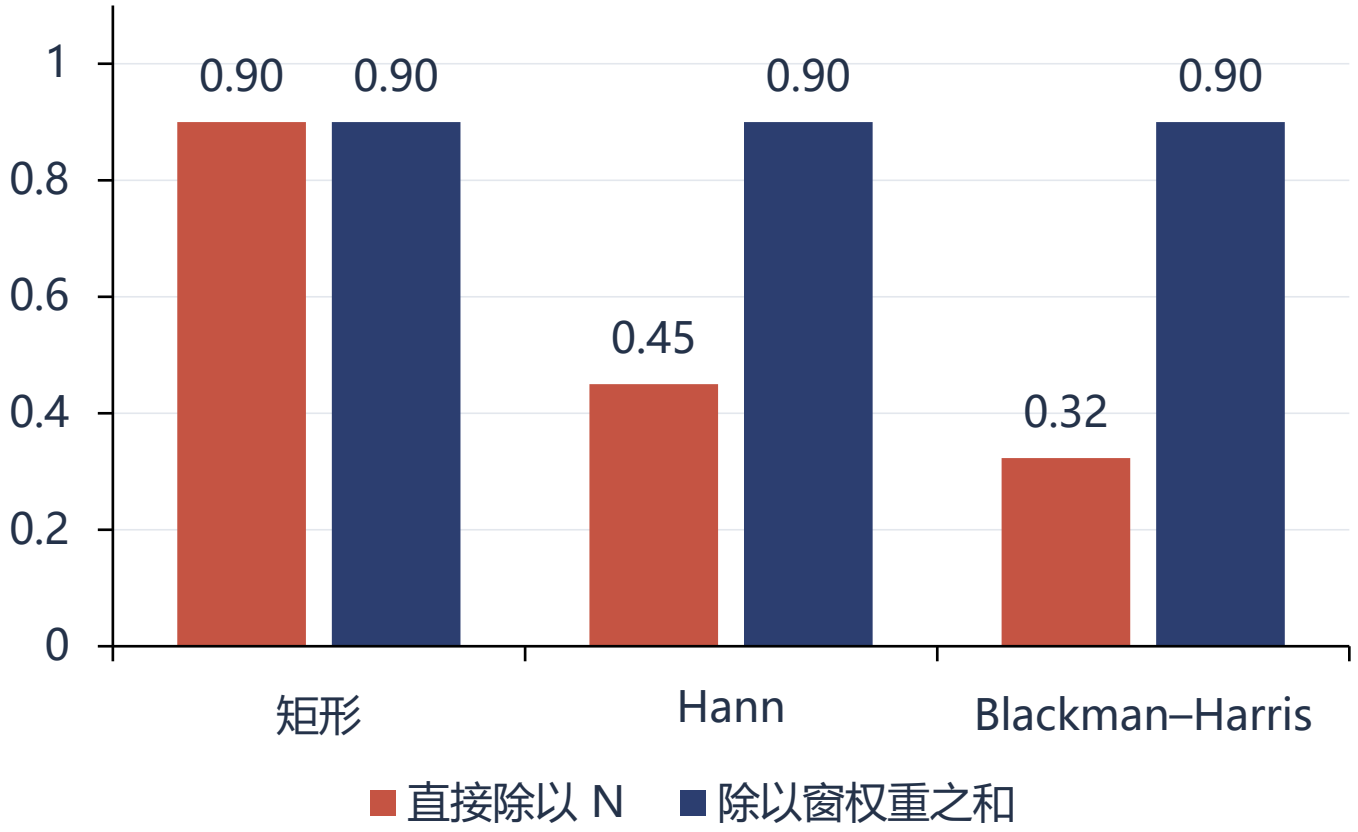
窗函数同时改变主瓣与旁瓣





窗函数	主瓣零点间宽度	最大旁瓣
矩形	2个频点间隔	−13.3 dB
Hann	4个频点间隔	−31.5 dB
Blackman–Harris	8个频点间隔	−92.0 dB

按窗的相干增益恢复峰值幅度



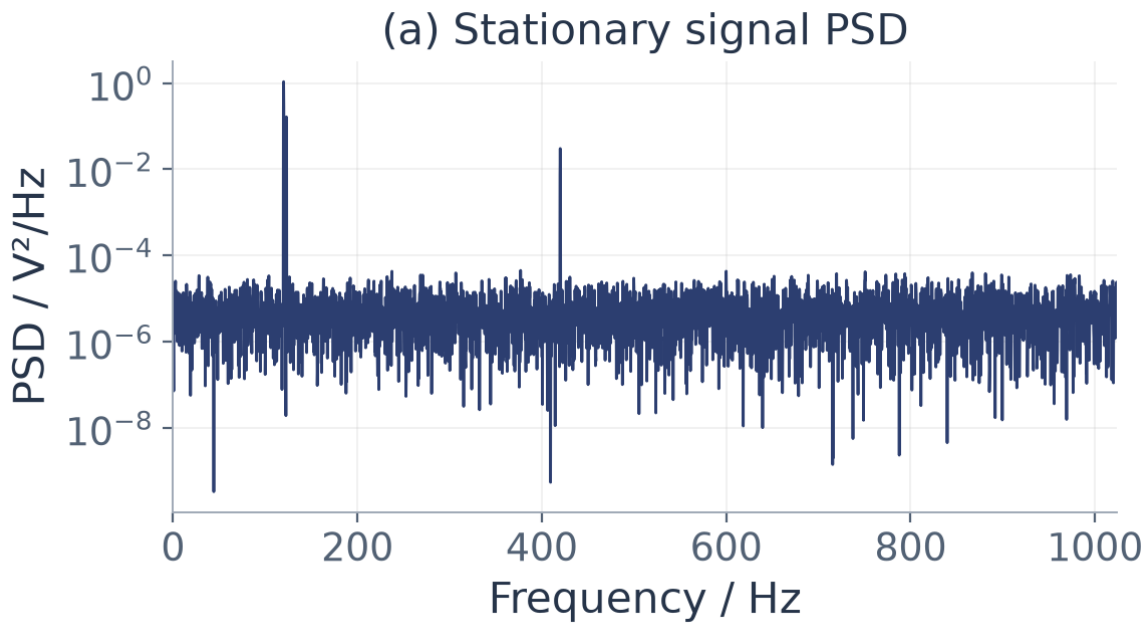
整频点正弦：峰值幅度0.9 V

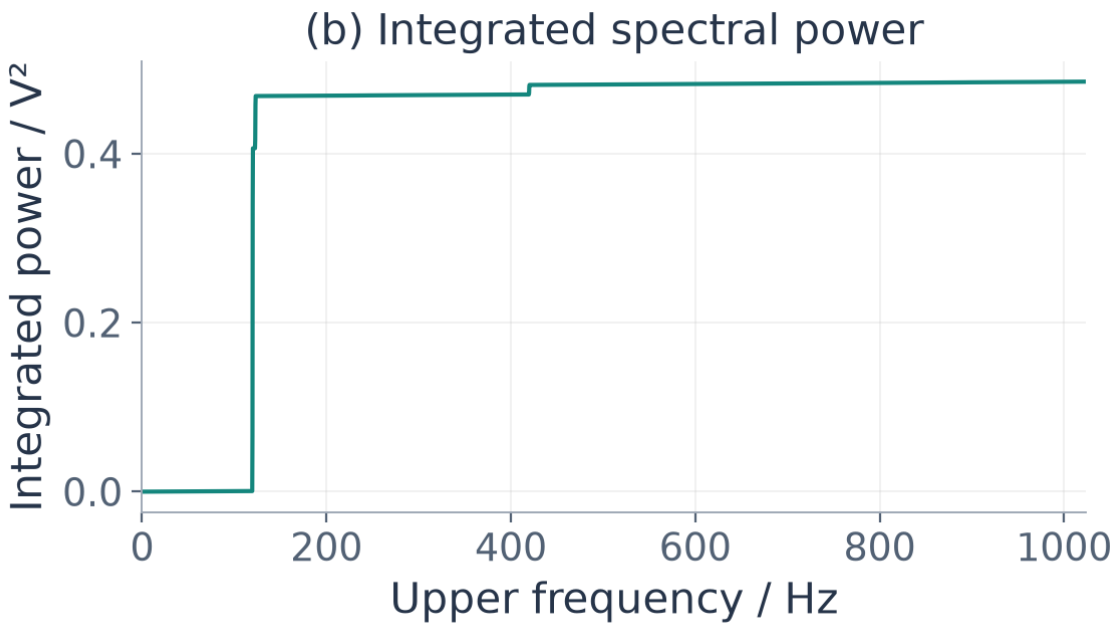
峰值幅度 = 2|Xw[k]| / Σw[n]
Hann 的平均权重为0.5，
除以窗权重总和恢复0.9 V。

正频率内部频点乘2；
DC（0 Hz）与Nyquist端点（$f_s/2$）保留单倍权重。

相干增益 = Σw[n]/N，表示窗对整频点正弦幅值的缩放。

功率谱密度积分回到信号均方值





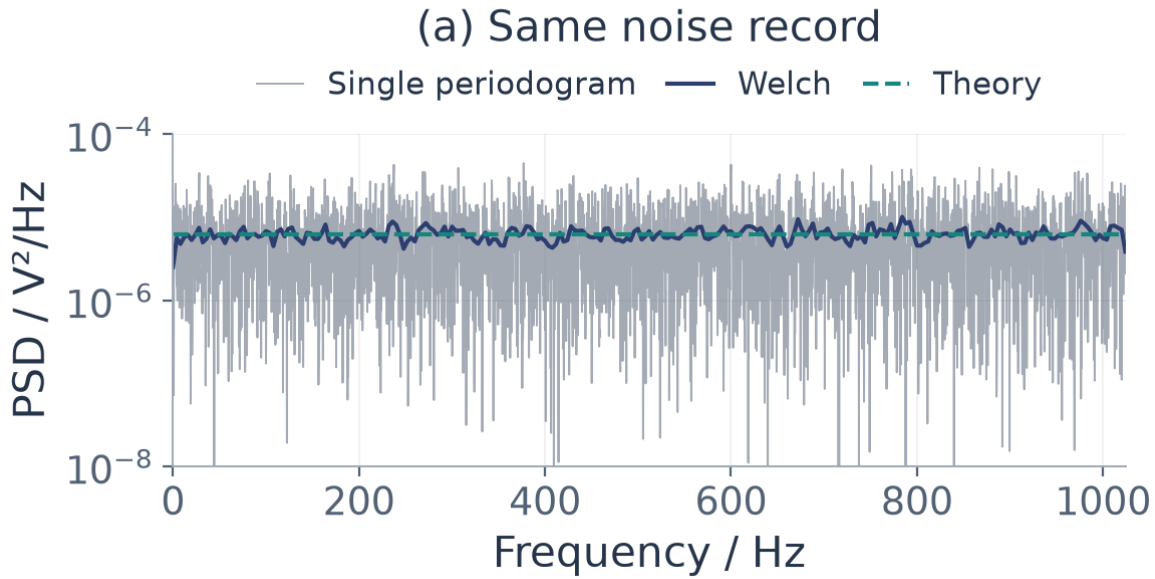
让纵轴单位回答功率问题

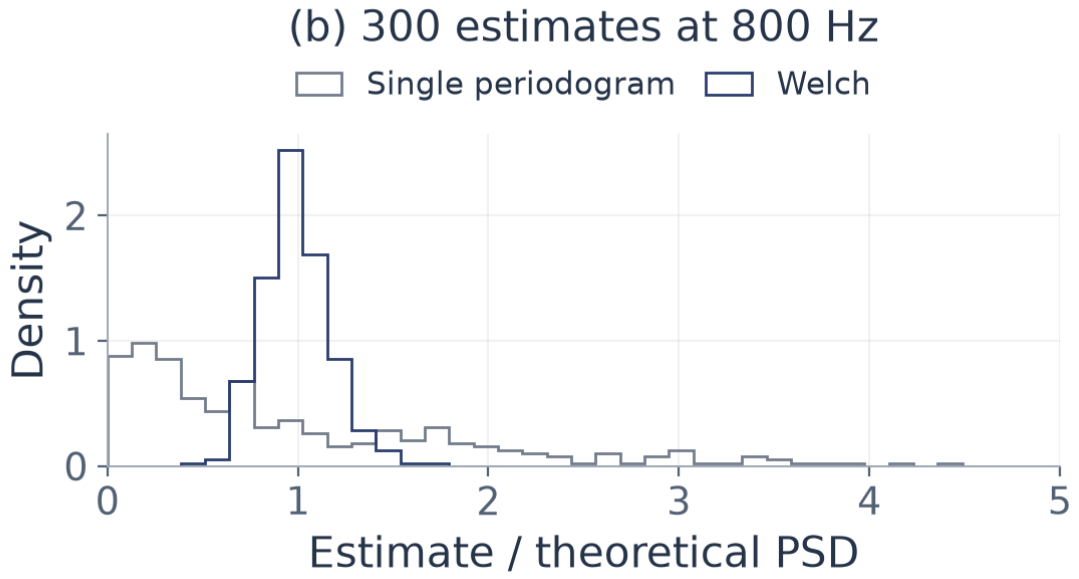
$PSD[k] = |Xw[k]|^2 / (f_s \Sigma w[n]^2)$
单边谱内部频点乘2，单位为V^2/Hz。
本页使用稳定分量与白噪声。

用积分做一致性检查

$\Sigma PSD[k]\cdot\Delta f = 0.4852\ V^2$。
时域核对：$\Sigma(x[n]w[n])^2 / \Sigma w[n]^2$；
两者相等，均方根约0.6966 V。

Welch 通过分段平均降低估计波动





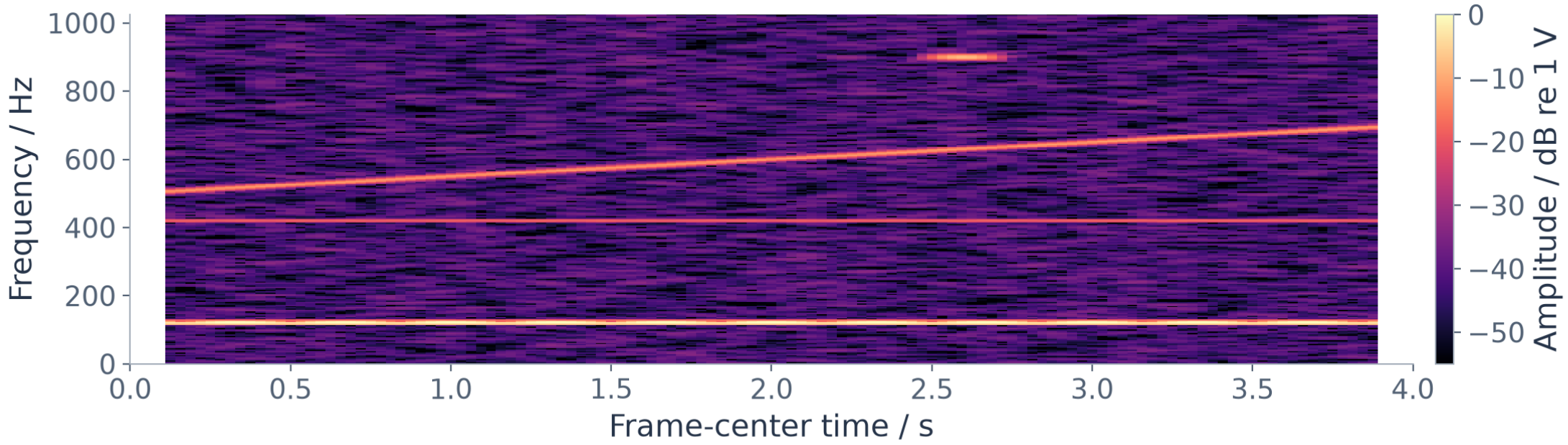
把长记录拆成多个重叠片段

8192点，片长512、重叠256点，
得到31段Hann窗谱并取平均。
分段长度同时改变频率分辨能力。

比较同一频率的重复估计

在800 Hz，300次独立噪声实验中，
标准差/均值从0.949降至0.180。
平均使随机噪声背景更稳定。

短时傅里叶变换定位频率随时间的变化



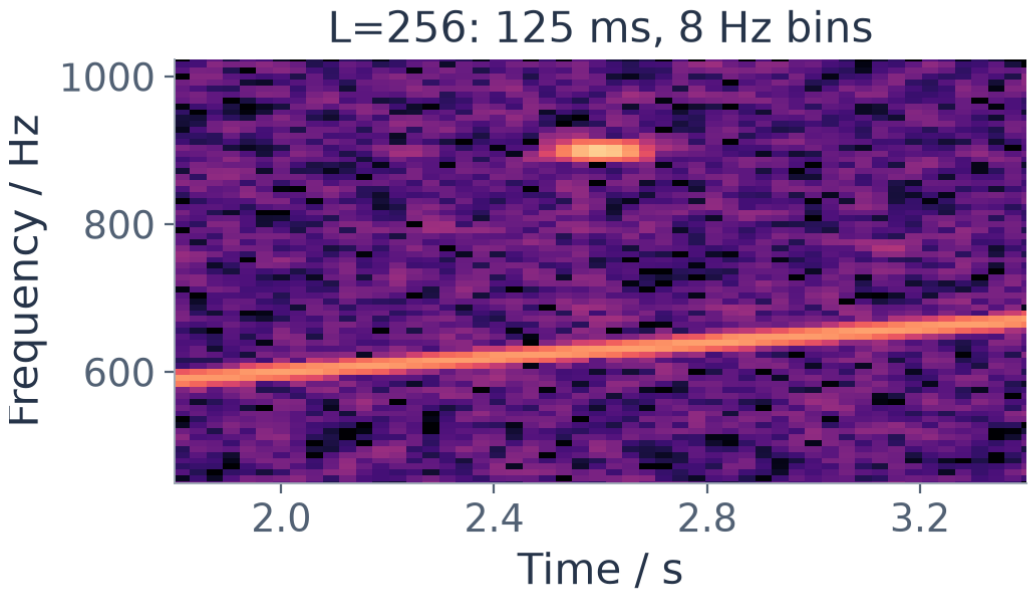
在滑动窗口内重复计算 FFT

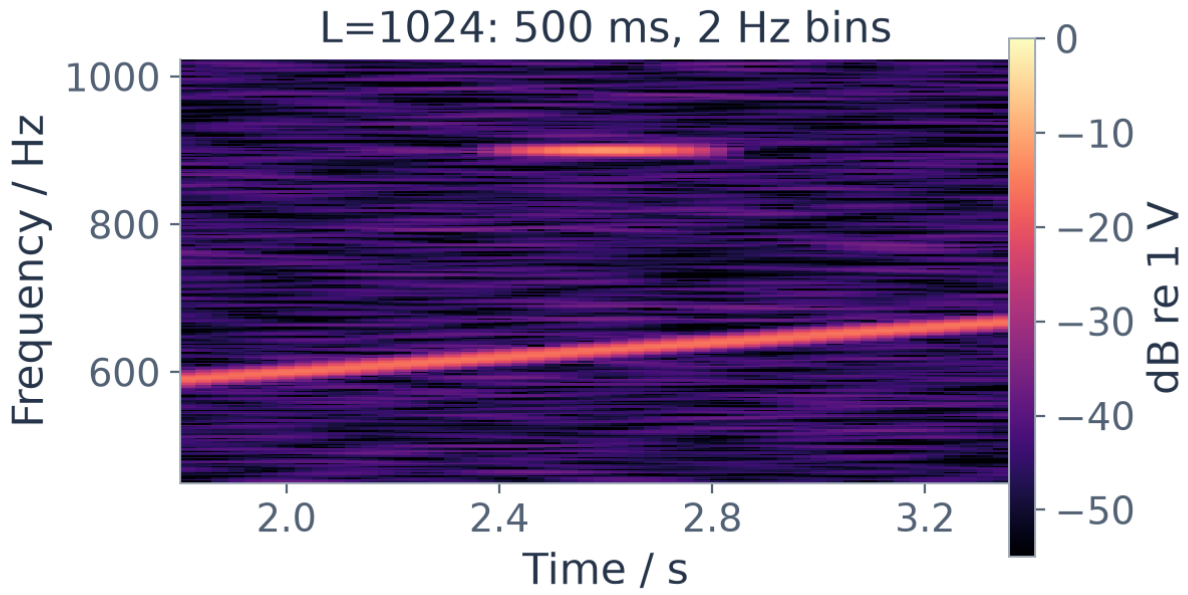
窗长512点（250 ms），步长64点。
纵轴为频率，横轴为窗口中心时刻，
颜色表示相干增益校准后的局部幅度。

将图中轨迹对应回构造信号

120/123 Hz合成低频亮带，420 Hz为水平线；
500→700 Hz扫频形成斜线，
900 Hz短时事件出现在约2.6 s。

窗长决定时域与频域的观察尺度





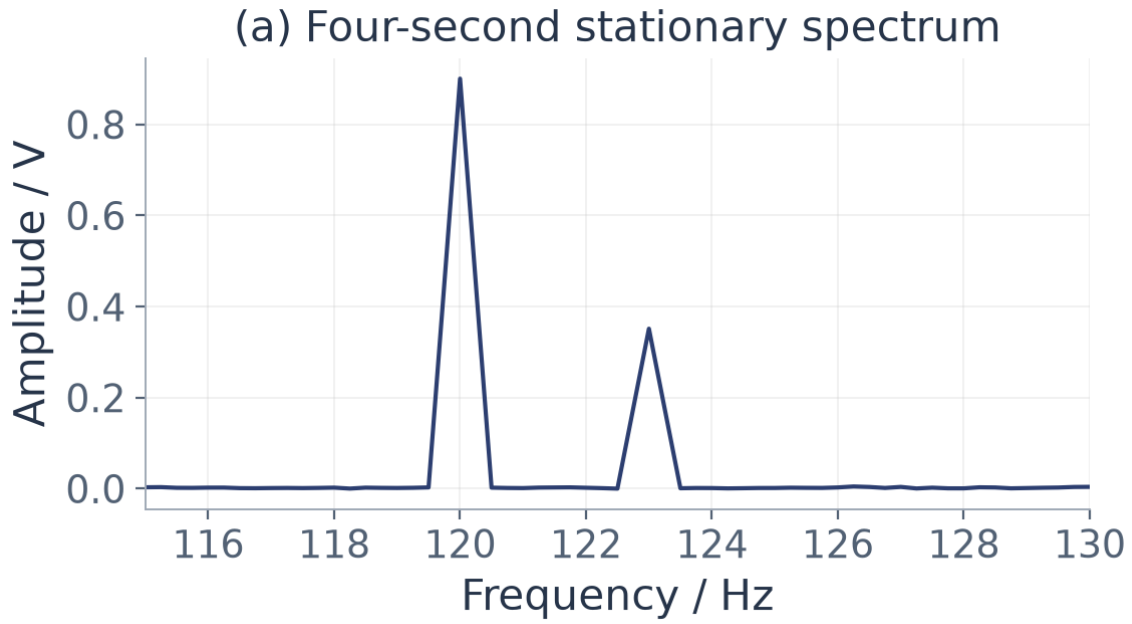
短窗：125 ms，8 Hz频率栅格

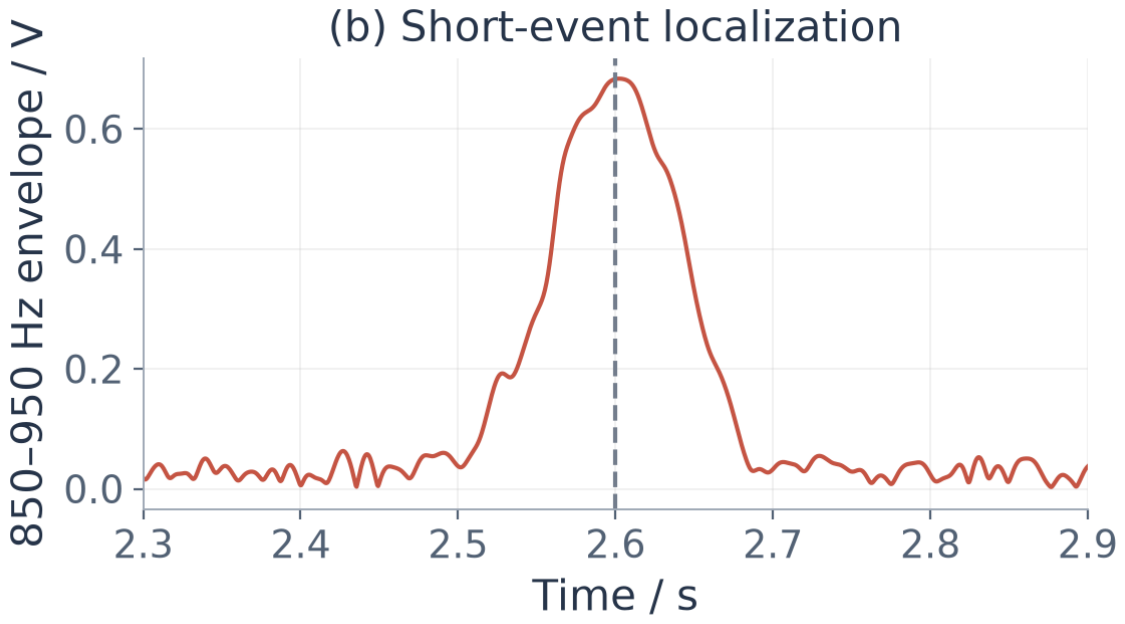
短时900 Hz事件在时间上更集中。
同一窗口包含的周期数更少，
相近频率的区分需要更长观察。

长窗：500 ms，2 Hz频率栅格

两图步长均为64点（31.25 ms）。
长窗使频率栅格更细，并在时间上平均；
窗长按所需现象选择。

将稳定峰与短时事件分开定位





稳定频率：使用完整4 s记录

Hann窗幅值谱给出120.00与123.00 Hz峰。
同一记录长度同时支撑
频率位置与近邻峰的区分。

短时事件：看目标频带包络

850–950 Hz带通后，用Hilbert变换
提取振幅包络。峰值位于2.602 s，
对应构造信号中心2.600 s。

把归一化与窗口参数写进代码

单边幅值谱

```
w = signal.get_window("hann", N)
X = np.fft.rfft(x * w)
f = np.fft.rfftfreq(N, 1/fs)
amp = np.abs(X) / w.sum()
amp[1:-1] *= 2
if N > 1 and N % 2:
    amp[-1] *= 2
```

x为实数采样序列，N=len(x)，fs为Hz；
先按分析目标去均值或去趋势。

分段功率谱与输入约定

```
f, psd = signal.welch(
    x, fs=fs, window="hann",
    nperseg=512, noverlap=256,
    detrend="constant",
    scaling="density")
power = psd.sum() * (f[1]-f[0])
```

np 为 NumPy，signal 为 scipy.signal。
本例的偶数N单独保留DC与Nyquist端点。

用可复算的检查锁定分析结果

检查	操作	本例结果
可逆性	irfft(rfft(x)) 与 x 比较	最大差 1.11×10^{-15} V
幅值	0.9 V整频点正弦，3种窗校准	最大差 1.33×10^{-15} V
功率	谱积分与窗加权时域均方比较	差 1.11×10^{-16} V^2
时间定位	包络峰与构造事件中心比较	2.602 s / 2.600 s

相同采样率、窗口、归一化与单位贯穿图表和代码，结论才对应同一个信号。

从问题出发，选定频谱分析的参数

要回答的问题	优先选择	核对依据
最高频率在哪里	采样率 + 抗混叠滤波	目标频段与Nyquist范围
两个峰能否分开	足够长的记录 + 合适窗	主瓣宽度与实际峰距
幅值和功率多大	相干增益校准 / PSD	V与V^2/Hz、功率积分
何时出现变化	STFT窗长/步长 + 频带包络	时间中心与目标频带

采样保留信息，窗口组织观察，归一化建立数值含义，复算检查连接整条分析链。

资料：NumPy FFT ｜ SciPy 频谱分析、Welch 与 ShortTimeFFT

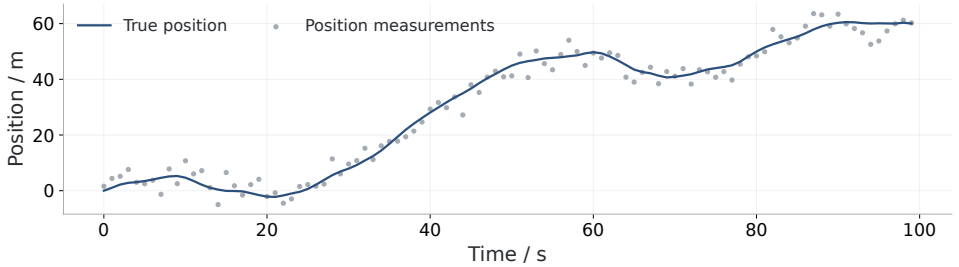
How Kalman Filters Track Noisy Motion

Prediction, uncertainty, and measurement updates

SKILL Application Test

A position sensor, a moving object, and a reusable estimator.

Track the motion behind noisy measurements



The observable signal

100 samples at 1-second intervals. The sensor measures position with standard deviation $3\,\mathrm{m}$.

The estimation task

Recover position and velocity, quantify their uncertainty, and continue through missing observations.

Represent the state and the measurement separately

State: what the system is doing

$$x_k = \begin{bmatrix} p_k \\ v_k \end{bmatrix}, \qquad \hat{x}_k = \begin{bmatrix} \hat{p}_k \\ \hat{v}_k \end{bmatrix}.$$

p_k: position in m.
v_k: velocity in m/s.
A hat marks an estimate. The subscript k identifies the sample time.

Measurement: what the sensor sees

$$z_k = Hx_k + e_k, \qquad H = \begin{bmatrix} 1 & 0 \end{bmatrix}.$$

The sensor observes position; velocity enters through the motion model over time.

$$e_k \sim \mathcal{N}(0, R), \qquad R = 9 \text{ m}^2.$$

Measurement variance is the square of its standard deviation.



A motion model carries information into the next step

Predict motion, allow acceleration

$$x_k = Fx_{k-1} + Ga_{k-1},$$

$$F = \begin{bmatrix} 1 & \Delta t \\ 0 & 1 \end{bmatrix}, \qquad G = \begin{bmatrix} \Delta t^2/2 \\ \Delta t \end{bmatrix}.$$

Random acceleration a_{k-1} changes both position and velocity over the interval.

Give model uncertainty a scale

$$a_{k-1} \sim \mathcal{N}(0, q), \qquad Q = qGG^T.$$

At $\Delta t = 1$ s and $q = 0.2\ (\mathrm{m/s^2})^2$,

$$Q = \begin{bmatrix} 0.05 & 0.10 \\ 0.10 & 0.20 \end{bmatrix}.$$

The tracking experiment uses independent acceleration and sensor noise across time.

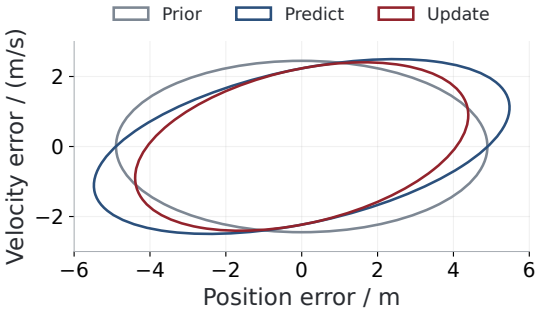
The units follow the state components

Q_{11} has units m^2, Q_{22} has units $(\mathrm{m/s})^2$, and the cross terms have units m^2/s.

Covariance describes the shape of uncertainty

$$P = \mathbb{E}[(x - \hat{x})(x - \hat{x})^T] = \begin{bmatrix} \sigma_p^2 & \sigma_{pv} \\ \sigma_{pv} & \sigma_v^2 \end{bmatrix}.$$

The diagonal entries describe component variances. The off-diagonal term links position error to velocity error.
A tilted ellipse means that a position correction also contains information about velocity.



This geometry starts from $P = \mathrm{diag}(4, 1)$, predicts for 1 s with $q = 0.04$, and updates with $R = 9$.

Contours enclose 95% of the joint Gaussian probability: $(x - \hat{x})^T P^{-1} (x - \hat{x}) = 5.991$.

First combine two scalar estimates

A fully worked position update

Superscripts $-$ and $+$ mean before and after the measurement update. Prior: $\hat{p}^- = 10$ m, $P^- = 4$ m^2.

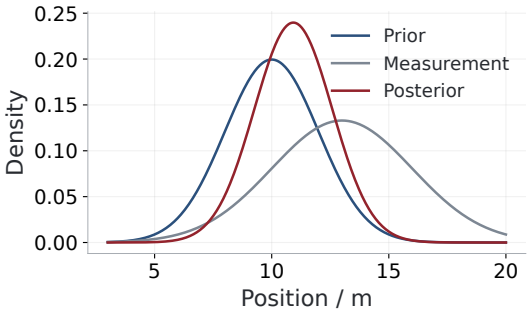
Measurement: $z = 13$ m, $R = 9$ m^2.

$$K = \frac{P^-}{P^- + R} = \frac{4}{13} \approx 0.308.$$

$$\hat{p}^+ = 10 + \tfrac{4}{13}(13 - 10) \approx 10.923 \text{ m}.$$

$$P^+ = (1 - K)P^- \approx 2.769 \text{ m}^2.$$

The estimate moves 0.923 m toward the observation.



The posterior combines the prior density and the measurement likelihood.

Predict both the state and its covariance

Propagation equations

$$\hat{x}_k^- = F\hat{x}_{k-1}^+, \qquad P_k^- = FP_{k-1}^+F^T + Q.$$

The superscript $-$ means before the current measurement; $+$ means after the update.
Prediction transports the previous uncertainty and adds process uncertainty.
In the position component,

$$P_{11}^- = P_{11}^+ + 2\Delta t P_{12}^+ + \Delta t^2 P_{22}^+ + Q_{11}.$$

One numerical step

Take $\Delta t = 1$ s and, for this calculation, $q = 0.04$.

$$\hat{x}^+ = \begin{bmatrix} 10 \\ 2 \end{bmatrix}, \quad P^+ = \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix}.$$

$$\hat{x}^- = \begin{bmatrix} 12 \\ 2 \end{bmatrix}, \quad P^- = \begin{bmatrix} 5.01 & 1.02 \\ 1.02 & 1.04 \end{bmatrix}.$$

Position and velocity become correlated because uncertain velocity affects the next position.

The innovation measures a prediction error

Measure the discrepancy

$$y_k = z_k - H\hat{x}_k^-, \qquad S_k = HP_k^-H^T + R.$$

y_k is the innovation. S_k includes uncertainty in the prediction and in the sensor.
The standardized innovation $y_k/\sqrt{S_k}$ puts errors on a comparable scale.

Continue the worked example

For $z = 13$ m, $H\hat{x}^- = 12$ m and $R = 9$ m^2,

$$y = 1 \text{ m}, \qquad S = 5.01 + 9 = 14.01 \text{ m}^2.$$

$$y/\sqrt{S} \approx 0.267.$$

A 1 m discrepancy is small relative to the combined predicted uncertainty.

The innovation standard deviation is $\sqrt{S} \approx 3.743$ m. Under the model, $y \sim \mathcal{N}(0, S)$; the 95% reference interval is ± 7.34 m.

One position observation can correct velocity too

The vector Kalman gain

$$K = P^{-}H^{T}S^{-1} = \frac{1}{14.01}\begin{bmatrix}5.01\\1.02\end{bmatrix} \approx \begin{bmatrix}0.3576\\0.0728\end{bmatrix}.$$

The first component converts position innovation into a position correction; the second converts it into a velocity correction.

Correct the prediction

$$\hat{x}^{+} = \hat{x}^{-} + Ky \approx \begin{bmatrix}12.3576\ \text{m}\\2.0728\ \text{m/s}\end{bmatrix}.$$

The positive cross-covariance makes an unexpectedly large position suggest a larger velocity.
K_p is dimensionless; K_v has units s^{-1}.

Same prediction, different sensor variance	R (m^2)	K_p	K_v (s^{-1})
Worked example	9	0.358	0.073
More precise sensor	1	0.834	0.170

Update uncertainty with the corrected state

Joseph form keeps the noise contributions explicit

$$A = I - KH, \qquad P^{+} = AP^{-}A^{T} + KRK^{T}.$$

Both terms are covariance contributions. This form is useful for preserving symmetry and positive-semidefinite structure in numerical implementations.

Before the observation

$$P^{-} = \begin{bmatrix} 5.01 & 1.02 \\ 1.02 & 1.04 \end{bmatrix}.$$

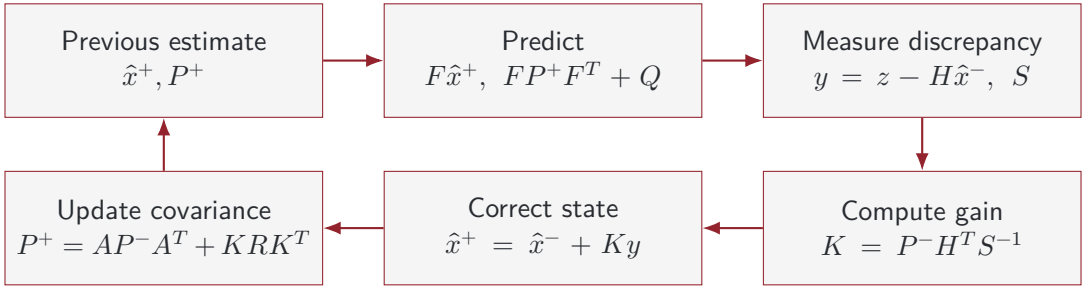
Position SD: $\sqrt{5.01} \approx 2.238\,\mathrm{m}$.

After the observation

$$P^{+} \approx \begin{bmatrix} 3.2184 & 0.6552 \\ 0.6552 & 0.9657 \end{bmatrix}.$$

Position SD: $\sqrt{3.2184} \approx 1.794\,\mathrm{m}$.

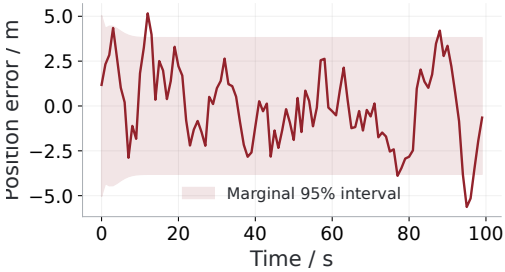
The estimator is a repeating predict-update loop

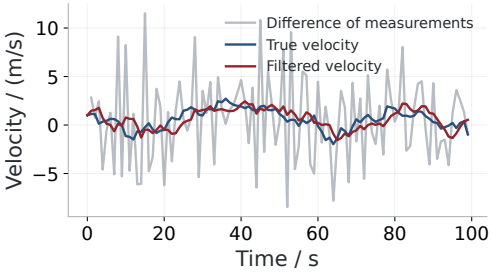


Each cycle carries forward a state vector and a covariance matrix. The innovation connects incoming data to the predicted state.

At a missing observation, retain the prediction and advance to the next timestamp.

The tracking run reduces position error





Position RMSE over steps 10–99

RMSE is the root mean square error.

Measurements: 3.133 m.

Filtered position: 2.143 m.

Read the uncertainty interval

Error means estimate minus truth. The shaded band is $\pm 1.96\sqrt{P_{11}}$; its width comes from the covariance recursion.

Velocity is inferred through motion over time

Direct differences amplify sensor noise

$$\tilde{v}_k = (z_k - z_{k-1})/\Delta t.$$

With independent sensor noise, its noise variance is $2R/\Delta t^2 = 18\ (\mathrm{m/s})^2$.
The Kalman state combines this temporal information with the motion model.

The same run, steps 10–99

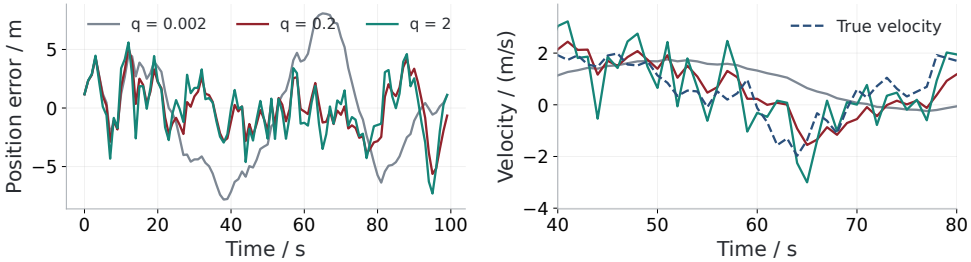
Velocity estimate	RMSE (m/s)
Measurement difference	4.204
Kalman state	0.848

Repeated positions constrain velocity because $HF = [1\ \Delta t]$ includes the velocity component.

Check that the hidden state can affect the observations

For $\Delta t \neq 0$, $\begin{bmatrix} H \\ HF \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & \Delta t \end{bmatrix}$ has rank 2: both state components are observable over time.

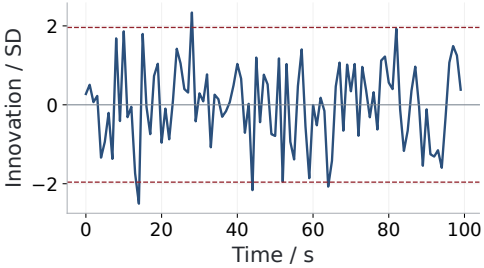
Process noise controls responsiveness

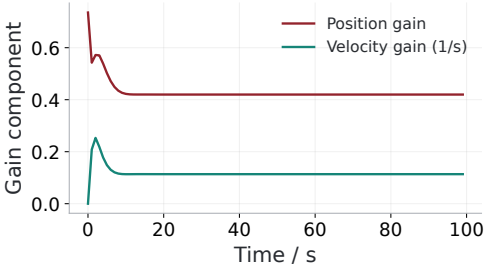


Assumed acceleration variance q	0.002	0.2	2.0
Position RMSE (m)	4.401	2.143	2.445
Velocity RMSE (m/s)	1.169	0.848	1.201

Small q makes the trajectory more persistent; larger q allows faster changes. Here the generating process uses $q = 0.2$.

Read the innovation and gain as diagnostics





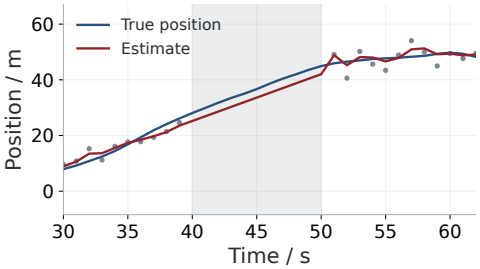
Innovation checks the prediction scale

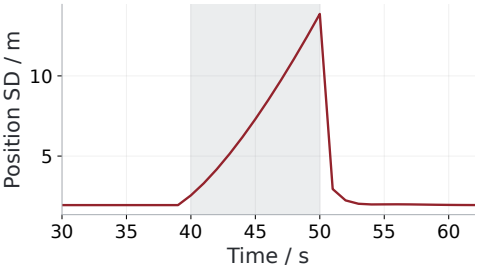
Read $y/\sqrt{S}$ against the dashed ± 1.96 reference. Persistent shifts can reveal a model or sensor change.

Gain records the evolving balance

The position and velocity gains settle as the covariance recursion approaches a stable regime.

Missing measurements increase uncertainty





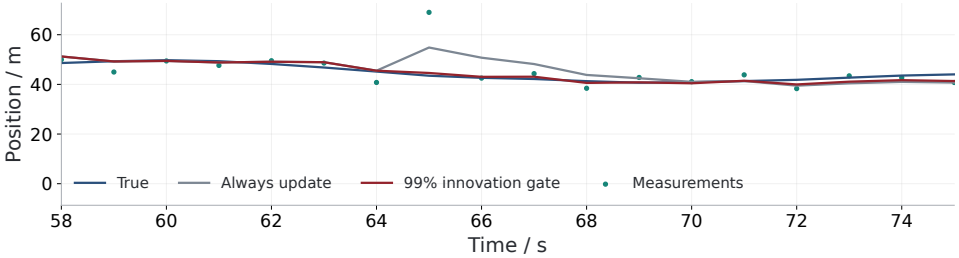
Prediction continues through the gap

Observations at steps 40–50 are withheld.
Apply the prediction equations at every step.

Position standard deviation

Step 39: 1.944 m. Step 50: 13.863 m.
The next observation reduces it to 2.944 m.

Use the innovation scale to screen a large outlier



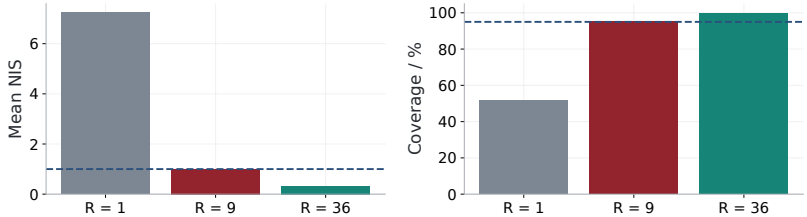
A normalized test

Normalized innovation squared: $\mathrm{NIS} = y^2/S$. For one Gaussian measurement, the 99% χ^2_1 threshold is 6.635.

An injected 30 m error at step 65

Its NIS is 38.42. The gate retains the prediction for this step; later observations resume correction.

Check covariance calibration across repeated trials



400 independent trials, 90 evaluated steps each; generating $q = 0.2$, $R = 9$.

Assumed R	1	9	36
Mean NIS	7.232	1.007	0.323
95% position-interval coverage	51.7%	95.1%	99.8%

Coverage is the fraction of true positions inside the estimated intervals. The matched model reaches 95.1%.

Translate the equations into a reusable update

```
def step(x, P, z, F, Q, H, R):
    xp = F @ x
    Pp = F @ P @ F.T + Q
    if z is None:
        return xp, Pp
    y = z - H @ xp
    S = H @ Pp @ H.T + R
    K = np.linalg.solve(
        S, H @ Pp).T
    x = xp + K @ y
    A = np.eye(len(x)) - K @ H
    P = A @ Pp @ A.T + K @ R @ K.T
    return x, (P + P.T) / 2
```

A clear array contract

x: length 2; P, F, Q: 2×2.
z: length 1; H: 1×2; R: 1×1.
Solve a linear system for the gain. Use Joseph form for the covariance, then remove numerical asymmetry.
The caller supplies elapsed-time-specific F and Q and handles any gating policy.
Earlier measurements are summarized by x and P; estimator storage stays fixed at the state size.

Build the estimator around a testable motion model

Design decision	What to establish
State and sensor	Name the physical variables, units and sensor mapping H.
Timing	Build $F(\Delta t)$ and $Q(\Delta t)$ from the motion assumptions.
Uncertainty	Set P_0, estimate R, and tune Q using errors and innovations.
Data handling	Continue prediction through gaps; define explicit outlier handling.
Validation	Compare estimates with truth and test interval coverage across trials.

The filter carries two things forward: an estimate of the state and a quantitative account of its uncertainty.

Further reading: Welch and Bishop, *An Introduction to the Kalman Filter*; MathWorks, Linear Kalman Filters.

How Indexes Accelerate Queries

From full scans to covering indexes

SKILL Application Test

A query, its data structure, and the work it performs

Start with one query and a fixed result

```
SELECT created_seq, amount_cents
FROM orders
WHERE tenant_id = 42
ORDER BY created_seq DESC
LIMIT 20;
```

200,000 synthetic orders

100 tenants; 1,980 matching rows
created_seq = creation order
amount_cents = amount in cents
payload = 64 bytes per row

Measure the work

Count SQLite virtual-machine (VM) instructions. Time full result retrieval: 3 warm-ups, then 11 executions. Report the median with a warm cache.

Keep the result fixed

Verify identical rows and order after every index change.
The access path is the experimental variable; the requested result stays fixed.

An index stores an additional order over the data

Row-ID order

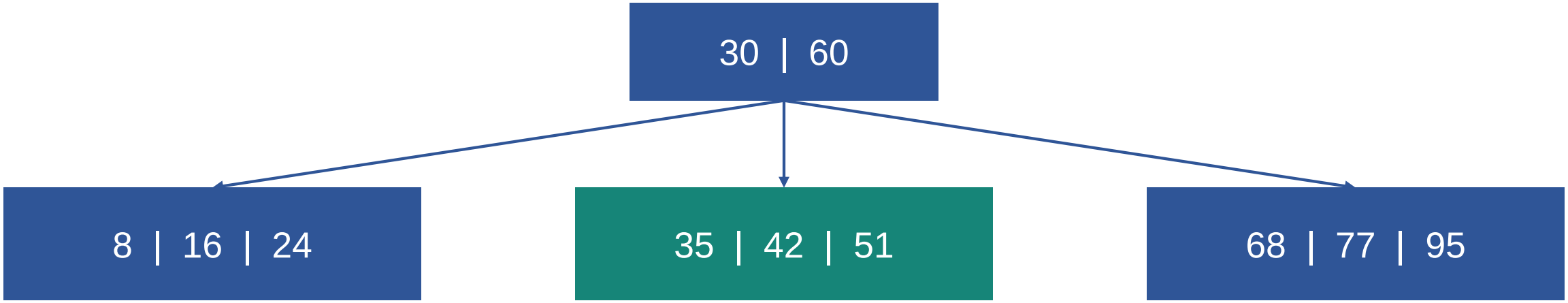
Row ID	Tenant	Sequence	Amount
17	42	16	8100
28	7	27	6400
66	42	65	2200
91	12	90	3500

Tenant-key order

Tenant	Row ID
7	28
12	91
42	17
42	66

These illustrative keys place tenant 42’s row IDs 17 and 66 next to each other.
The row IDs locate the corresponding records in the original table.

A B-tree routes the search through a few pages



Search key 42

Compare separator keys, descend into the middle range, and find the matching entry.

30 < 42 < 60: follow the middle child, then locate 42 among 35, 42 and 51.

Illustrative keys; each node represents a storage page.

High fan-out keeps the tree shallow

Index page type	Pages	Entries	Storage
Internal	6	845	24 KiB
Leaf	846	199,155	3.305 MiB

Fan-out is the number of child links in an internal page.

A three-level structure

The measured index has a root,
an internal level and leaf pages.
Its longest path contains 2 edges.

Work follows the access path

A lookup routes to a small region.
A full scan visits the index or table
sequentially across its entries.

Without an index, filter and sort the table



EXPLAIN QUERY PLAN reports:

```
SCAN orders
USE TEMP B-TREE FOR ORDER BY
```

Baseline: 617,851 virtual-machine instructions to retrieve the 20 rows.

A tenant index removes most filtering work

```
CREATE INDEX idx ON orders(tenant_id);
```



The equality constraint is searchable

SEARCH orders USING INDEX idx (tenant_id=?)

The requested order remains separate

A temporary B-tree still supplies ordering. Measured work falls to 23,792 instructions.

A composite index groups and orders together

```
CREATE INDEX idx ON orders(tenant_id, created_seq);
```

Tenant	Created sequence	Illustrative key order
41	199800	Previous tenant
42	100	Start of tenant 42
42	95000	Same tenant, increasing sequence
42	199943	Recent row inside the same range
43	15	Next tenant

Equality on tenant selects one contiguous range; sequence orders the entries within it.

An ordered index can stop after the first 20 matches




```
SEARCH orders USING INDEX idx (tenant_id=?)
```

The order comes from the index

The plan no longer reports
USE TEMP B-TREE FOR ORDER BY.

Early termination changes the workload

Measured instructions: 150.
The query retrieves a small tail of
one tenant’s ordered range.

A covering index supplies every requested value

```
CREATE INDEX idx ON orders(tenant_id, created_seq, amount_cents);
```

Index field	Role in this query
tenant_id	Equality search
created_seq	Requested order and output
amount_cents	Output without a table lookup

The plan reports USING COVERING INDEX; measured work is 129 instructions.

Column order determines the searchable prefix

```
SELECT sum(amount_cents)
FROM orders INDEXED BY idx
WHERE tenant_id = 42
  AND created_seq BETWEEN 80000 AND 119999
```

Index order	Searchable region	VM instructions
tenant_id, created_seq	One tenant within the sequence range	1,991
created_seq, tenant_id	Sequence range across tenants	200,808

These paths are selected with INDEXED BY to isolate key order in the experiment.

Coverage depends on the requested output columns

Selected values	Plan	Table records needed
created_seq, amount_cents	USING COVERING INDEX	No
created_seq, amount_cents, payload	USING INDEX	Yes

The index already holds

Tenant, sequence and amount are inside the composite key.

The extra value lives in the table

Adding payload requires row lookups.
This version executes
170 instructions.

A partial index represents a useful subset

```
CREATE INDEX idx ON orders(tenant_id, created_seq, amount_cents)
WHERE status = 'open';
```

Index	Rows represented	Index storage
Partial active index	20,045	0.336 MiB
Full: (status, tenant_id, created_seq, amount_cents)	200,000	4.641 MiB

The query includes status = 'open', so the partial index contains every eligible row.

An expression can change the usable search constraint

Predicate	Observed plan	VM instructions
tenant_id = 42	SEARCH ... (tenant_id=?)	5,953
tenant_id + 0 = 42	SCAN ... COVERING INDEX	801,994

Same answer, different access path

Both queries count the same 1,980 matching rows.

Make the intended access visible

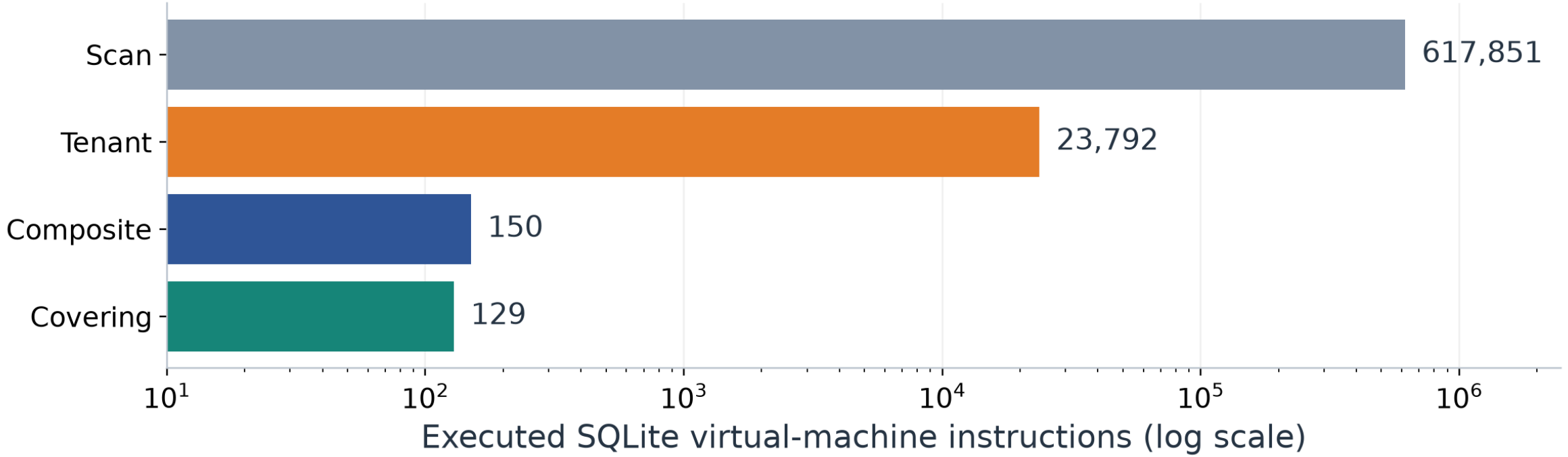
Keep the indexed equality directly searchable, or design an index for the actual expression being queried.

Read the query plan as a physical route

Plan detail	What to look for
SCAN	Traversal through the selected table or index
SEARCH ... (tenant_id=?)	A constrained lookup into a key range
USING COVERING INDEX	Requested fields supplied directly by the index
USE TEMP B-TREE FOR ORDER BY	A separate structure supplies sorting

Pair the plan with returned rows and measured work; each explains a different part of execution.

The same result can require radically different work



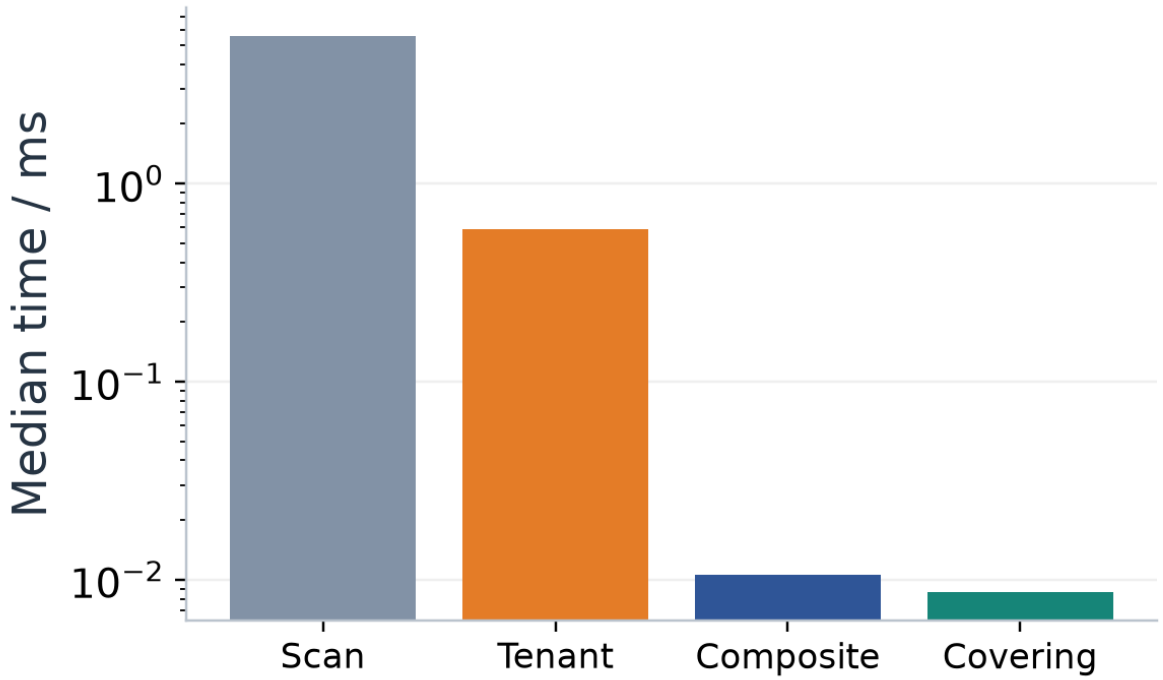
Progressively remove work

Search a tenant, preserve the order, then supply the output columns inside the index.

From the baseline to coverage

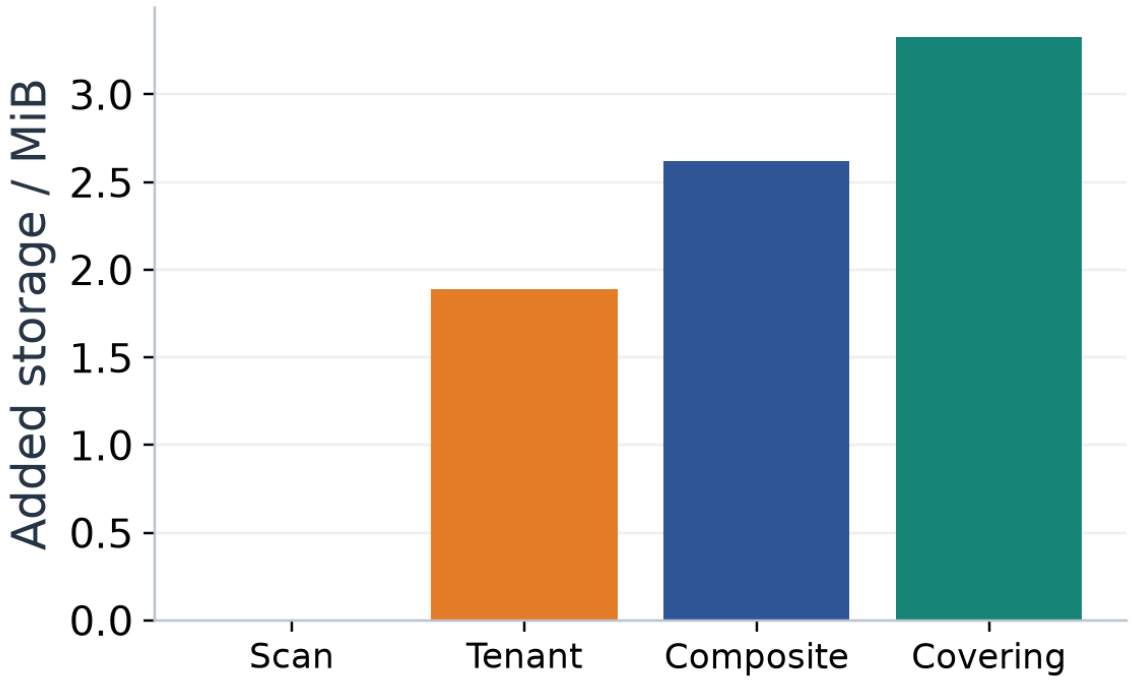
617,851 → 129 instructions.
All four cases return identical rows in identical order.

Latency and storage reveal different costs



Median warm-cache times

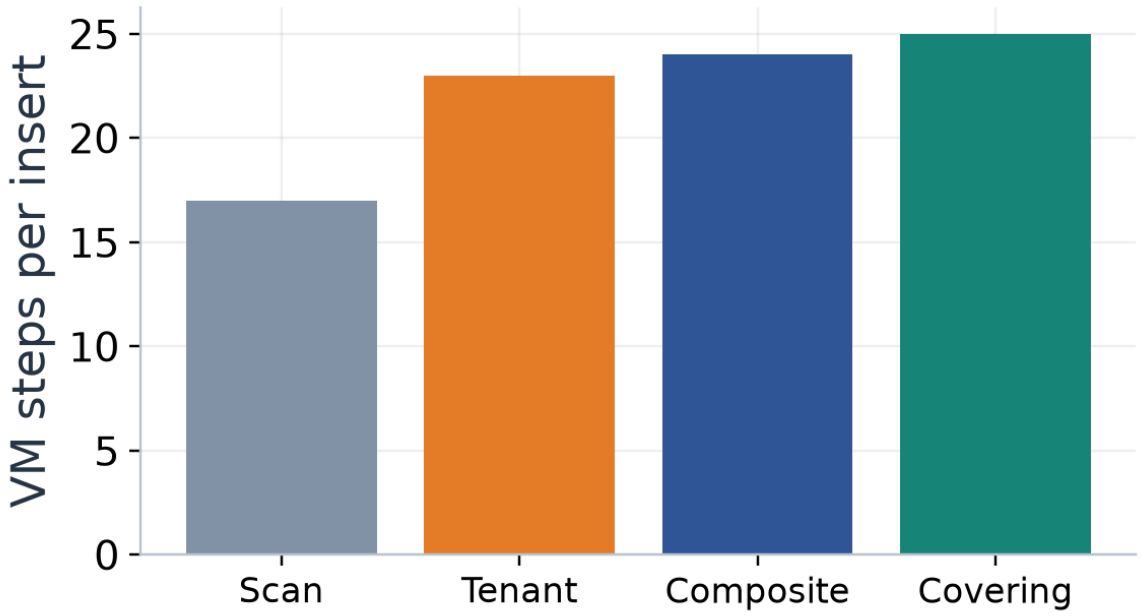
SQLite 3.53.1; in-memory database.
200,000 rows; 11 timed executions
after 3 warm-ups per query.

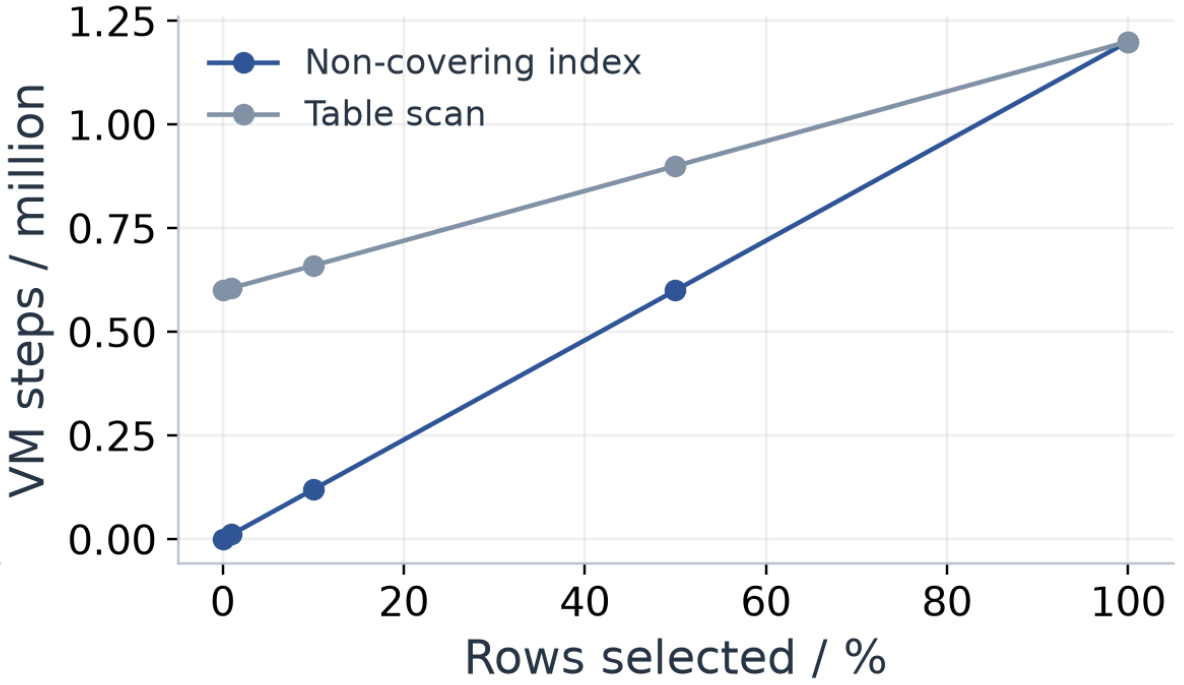


The added order takes space

Each index stores another key order.
The covering form also carries
amount_cents for direct output.

Index maintenance and broad reads also consume work





Each insert maintains every index

The left panel counts instructions for the same 5,000-row transaction, rolled back after measurement.

A broad query touches more records

The right panel forces each access path and fetches payload from the table. At 100%, both traverse the full dataset.

Build indexes through a repeatable query experiment

Step	Concrete action
1 Define the query	Fix predicates, ordering, output columns and row limit.
2 Capture the baseline	Save EXPLAIN QUERY PLAN, results and work measurements.
3 Choose the key	Place equality/range/order fields to match the access path.
4 Measure the trade-off	Check read work, storage and write maintenance.
5 Refresh statistics	Use ANALYZE or PRAGMA optimize, then inspect the plan again.

For each query, compare candidate indexes against that query’s fixed baseline result.

Choose the index that matches the work to be done

Query requirement	Index property to evaluate
Selective equality	A usable leading key
Ordered limited results	Order after equality-constrained key fields
Output without row lookups	Coverage of every selected value
Active subset	A partial-index predicate implied by the query
Frequent writes or broad reads	Maintenance cost, storage and actual access work

An index helps by narrowing, ordering or supplying data along the query's actual path.

SQLite documentation: Query Planning · EXPLAIN QUERY PLAN · Partial Indexes